## Summary


1. There are four major morphological groups of phages: tailless icosahedral phages, phages with contractile tails, those with noncontractile tails, and filamentous phages (**figure 17.1**).
2. The lytic cycle of virulent bacteriophages is a life cycle that ends with host cell lysis and virion release.
3. The phage life cycle can be studied with a one-step growth experiment that is divided into an initial eclipse period within the latent period, and a rise period or burst (**figure 17.2**).
4. The life cycle of the dsDNA T4 phage of *E. coli* is composed of several phases. In the adsorption phase the phage attaches to a specific receptor site on the bacterial surface. This is followed by penetration of the cell wall and insertion of the viral nucleic acid into the cell (**figure 17.3**).
5. Transcription of T4 DNA first produces early mRNA, which directs the synthesis of the protein factors and enzymes required to take control of the host and manufacture phage nucleic acids (**figure 17.5**).
6. T4 DNA contains hydroxymethylcytosine (HMC) in place of cytosine, and glucose is often added to the HMC to protect the phage DNA from attack by host restriction enzymes.
7. T4 DNA replication produces concatemers, long strands of several genome copies linked together.
8. Late mRNA is produced after DNA replication and directs the synthesis of capsid proteins, proteins involved in phage assembly, and those required for cell lysis and phage release.
9. Complete virions are assembled immediately after the separate components have been constructed. This is a self-assembly process, but does require participation of the bacterial membrane and a few extra proteins.
10. T4, $\phi$X174, and many other phages are released upon lysis of the host cell.
11. The replication of ssDNA phages proceeds through the formation of a double-stranded replicative form (RF) (**figure 17.11**). The filamentous ssDNA phages are continually released without host cell lysis.
12. When ssRNA bacteriophage RNA enters a bacterial cell, it acts as a messenger and directs the synthesis of RNA replicase, which then produces double-stranded replicative forms and, subsequently, many +RNA copies (**figure 17.13**).
13. The $\phi$6 phage is the only dsRNA phage known. It is also unusual in having a membranous envelope.
14. Temperate phages, unlike virulent phages, often reproduce in synchrony with the host genome to yield a clone of virus-infected cells. This relationship is lysogeny, and the infected cell is called a lysogen. The latent form of the phage genome within the lysogen is the prophage (**figure 13.18**).
15. Lysogeny is reversible, and the prophage can be induced to become active again and lyse its host.
16. A temperate phage may induce a change in the phenotype of its host cell that is not directly related to the completion of its life cycle. Such a change is called a conversion.
17. Two of the first proteins to appear after infection with lambda are the lambda repressor and the cro protein. The lambda repressor blocks the transcription of both cro protein and those proteins required for the lytic cycle, while the cro protein inhibits transcription of the lambda repressor gene (**figure 17.18**).
18. There is a race between synthesis of lambda repressor and that of the cro protein. If the cro protein level rises high enough in time, lambda repressor synthesis is blocked and the lytic cycle initiated; otherwise, all genes other than the lambda repressor gene are repressed and the cell becomes a lysogen.
19. The final step in prophage formation is the insertion or integration of the lambda genome into the *E. coli* chromosome; this is catalyzed by a special integrase enzyme (**figure 17.20**).
20. Several environmental factors can lower repressor levels and trigger induction. The prophage becomes active and makes an excisionase protein that causes the integrase to reverse integration, free the prophage, and initiate a lytic cycle.


## Key Terms

## Questions for Thought and Review

1. Explain why the T4 phage genome is circularly permuted.
2. Can you think of a way to simplify further the genomes of the ssRNA phages MS2 and Qβ? Would it be possible to eliminate one of their genes? If so, which one?
3. No temperate RNA phages have yet been discovered. How might this absence be explained?
4. How might a bacterial cell resist phage infections? Give those mechanisms mentioned in the chapter and speculate on other possible strategies.

## Critical Thinking Questions

1. The choice between lysogeny and lysis is influenced by many factors. How would external conditions such as starvation or crowding be "sensed" and communicated to the transcriptional machinery and influence this choice?
2. If you were a doctor charged with curing a bacterium of its viral infection, what target would you choose for "chemotherapy" and why?
3. We don't know exactly how double-stranded RNA is replicated. Propose two possible models, and design experiments that would distinguish between them.
4. The most straightforward explanation as to why the endolysin of T4 is expressed so late in infection is that its promoter is recognized by the gp55 alternative sigma factor. Propose a different explanation.

## Additional Reading

Chapter 16 references also should be consulted, particularly the introductory and advanced texts.

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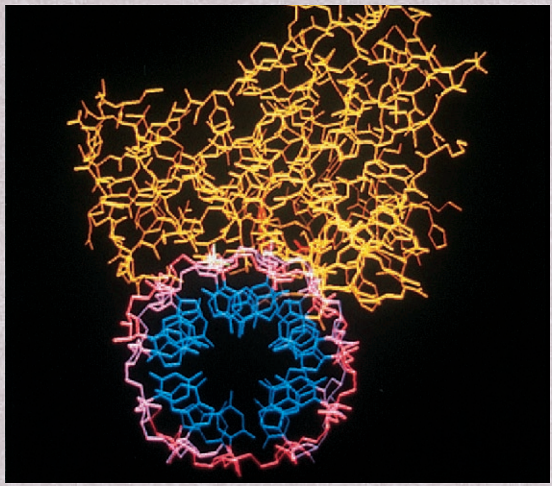
CHAPTER 18

# The Viruses: Viruses of Eucaryotes



A model of the ribonuclease H component of reverse transcriptase that is complexed with an RNA-DNA hybrid (protein in yellow, DNA backbone in lavender, RNA in pink, bases in blue).

## Outline



## Concepts

1. Although the details differ, animal virus reproduction is similar to that of the bacteriophages in having the same series of phases: adsorption, penetration and uncoating, replication of virus nucleic acids, synthesis and assembly of capsids, and virus release.
2. Viruses may harm their host cells in a variety of ways, ranging from direct inhibition of DNA, RNA, and protein synthesis to the alteration of plasma membranes and formation of inclusion bodies.
3. Not all animal virus infections have a rapid onset and relatively short duration. Some viruses establish long-term chronic infections; others are dormant for a while and then become active again. Slow virus infections may take years to develop.
4. Cancer can be caused by a number of factors, including viruses. Viruses may bring oncogenes into a cell, carry promoters that stimulate a cellular oncogene, or in other ways transform cells into tumor cells.
5. Plant viruses are responsible for many important diseases but have not been intensely studied due to technical difficulties. Most are RNA viruses. Insects are the most important transmission agents, and some plant viruses can even multiply in insect tissues before being inoculated into another plant.
6. Members of at least seven virus families infect insects; the most important belong to the *Baculoviridae, Reoviridae,* or *Iridoviridae.* Many insect infections are accompanied by the formation of characteristic inclusion bodies. A number of these viruses show promise as biological control agents for insect pests.

## Concepts *(continued)*

7. Infectious agents simpler than viruses also exist. Viroids are short strands of infectious RNA responsible for several plant diseases. Prions or virinos are somewhat mysterious proteinaceous particles associated with certain degenerative neurological diseases in humans and livestock.

*The Virus*
*Observe this virus: think how small*
*Its arsenal, and yet how loud its call;*
*It took my cell, now takes your cell,*
*And when it leaves will take our genes as well.*
*Genes that are master keys to growth*
*That turn it on, or turn it off, or both;*
*Should it return to me or you*
*It will own the skeleton keys to do*
*A number on our tumblers; stage a coup.*

—*Michael Newman*

In chapter 17 the bacteriophages are introduced in some detail because they are very important to the fields of molecular biology and genetics, as well as to virology. The present chapter focuses on viruses that use eucaryotic organisms as hosts. Although plant and insect viruses are discussed, particular emphasis is placed on animal viruses since they are very well studied and the causative agents of so many important human diseases. The chapter closes with a brief summary of what is known about infectious agents that are even simpler in construction than viruses: the viroids and prions.

The chapter begins with a discussion of animal viruses. These viruses are not only of great practical importance but are the best studied of the virus groups to be described in this chapter.

## 18.1 Classification of Animal Viruses

When microbiologists first began to classify animal viruses, they naturally thought in terms of features such as the host preferences of each virus. Unfortunately not all criteria are equally useful. For example, many viruses will infect a variety of animals, and a particular animal can be invaded by several dissimilar viruses. Thus virus host preferences lack the specificity to distinguish precisely among different viruses. Modern classifications are primarily based on virus morphology, the physical and chemical nature of virion constituents, and genetic relatedness.

Morphology is probably the most important characteristic in virus classification. Animal viruses can be studied with the transmission electron microscope while still in the host cell or after release. As mentioned in chapter 16, the nature of virus nucleic acids is also extremely important. Nucleic acid properties such as the general type (DNA or RNA), strandedness, size, and segmentation are all useful. Genetic relatedness can be estimated by techniques such as nucleic acid hybridization, nucleic acid and protein sequencing, and by determining the ability to undergo recombination.

A brief diagrammatic description of DNA and RNA animal virus classification is presented in **figures 18.1** and **18.2. Figure 18.3** summarizes pictorially much of the same material.

1. List the most important characteristics used in identifying animal viruses (figures 18.1, 18.2, and 18.3). What other properties can be used to establish the relatedness of different viruses?

## 18.2 Reproduction of Animal Viruses

The reproduction of animal viruses is very similar in many ways to that of phages. Animal virus reproduction may be divided into several stages: adsorption, penetration and uncoating, replication of virus nucleic acids, synthesis and assembly of virus capsids, and release of mature viruses. Each of these stages will be briefly described.

### Adsorption of Virions

The first step in the animal virus cycle is adsorption to the host cell surface. This occurs through a random collision of the virion with a plasma membrane receptor site protein, frequently a glycoprotein (a protein with carbohydrate covalently attached). Because the capacity of a virus to infect a cell depends greatly on its ability to bind to the cell, the distribution of these receptor proteins plays a crucial role in the tissue and host specificity of animal viruses. For example, poliovirus receptors are found only in the human nasopharynx, gut, and spinal cord anterior horn cells; in contrast, measles virus receptors are present in most tissues. The dissimilarity in the distribution of receptors for these two viruses helps explain the difference in the nature of polio and measles.

The specific host cell receptor proteins to which viruses attach vary greatly but they are always surface proteins necessary to the cell. As will be discussed shortly, viruses often enter cells by endocytosis. They trick the host cell by attaching to surface molecules that are normally taken up by endocytosis. Thus they are passively carried into the cell. These host cell surface proteins usually are receptors that bind hormones and other important molecules essential to the cell's function and role in the body (**table 18.1**).

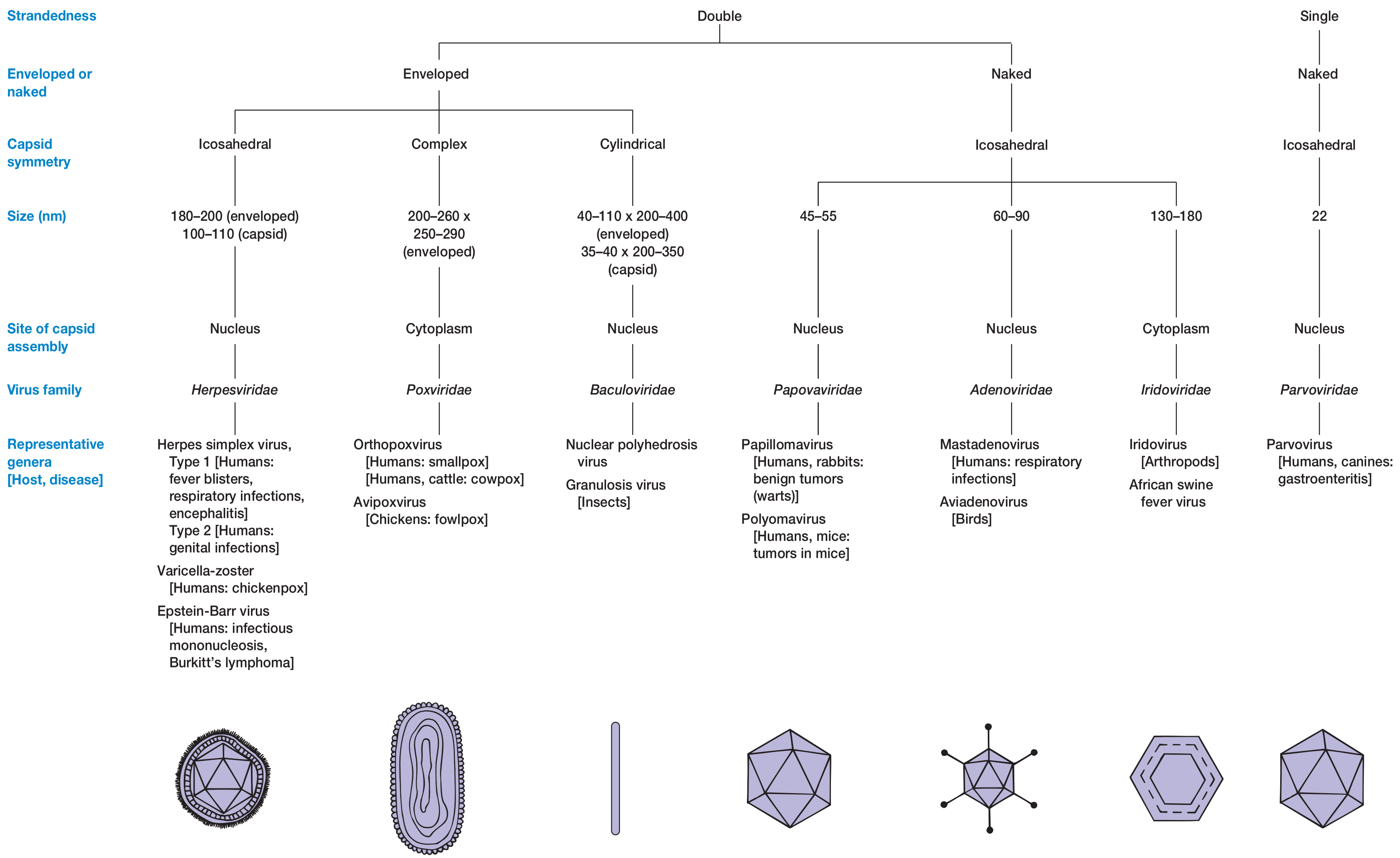



**Figure 18.1** **The Taxonomy of DNA Animal Viruses.**

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Strandedness** | Double | Single | | | | | | | | | |
| **Single-strand sense** | | Positive | | | | | Negative | | | | |
| | | | | | | DNA intermediate during replication | | | | | |
| **Enveloped or naked** | Naked | Enveloped | Enveloped | Naked | Naked | Enveloped | Enveloped | Enveloped | Enveloped | Enveloped | Enveloped |
| **Capsid symmetry** | Icosahedral | Icosahedral | Helical | Icosahedral | Icosahedral | Icosahedral | Helical | Helical | Helical | ? | Helical |
| **Size (nm)** | 75–80 | 40–75 (enveloped) 25–35 (capsid) | 80–160 (enveloped) 14–16 (capsid diameter) | 22–30 | 35–40 | 80–100 (enveloped) | 80–120 (enveloped) 9 (capsid diam.) | 70–80 x 130–240 (bullet-shaped) | 90–100 (enveloped) | 50–300 | 125–250 (enveloped) 18 (capsid diam.) |
| **Site of capsid assembly** | Cytoplasm | Cytoplasm | Cytoplasm | Cytoplasm | Cytoplasm | Cytoplasm | Cytoplasm | Cytoplasm | Cytoplasm | Cytoplasm | Cytoplasm |
| **Virus family** | *Reoviridae* | *Togaviridae* | *Coronaviridae* | *Picornaviridae* | *Calciviridae* | *Retroviridae* | *Orthomyxoviridae* | *Rhabdoviridae* | *Bunyaviridae* | *Arenaviridae* | *Paramyxoviridae* |
| **Representative genera and groups [Host, disease]** | Orbivirus [Humans: encephalitis]; Rotavirus [Humans: diarrhea]; Cypovirus [Insects: cytoplasmic polyhedrosis virus | Alphavirus [Humans: encephalitis]; Flavivirus [Humans: yellow fever, dengue]; Rubivirus [Humans: rubella]; Pestivirus [Pigs: hog cholera] | Infectious bronchitis virus [Humans: upper respiratory infection] | Enterovirus [Humans: polio]; Rhinovirus [Humans: common cold]; Hepatovirus [Humans: hepatitis A] | Vesicular exanthema of swine; Norwalk virus; Hepatitis E virus | Oncornavirus C [Birds, mice: sarcomas, leukemias]; Human T-cell leukemia virus; Human immunodeficiency virus | Influenza virus [Humans, swine] | Lyssavirus [Warm-blooded animals: rabies] | California encephalits virus [Humans] | Lassa virus [Humans: hemorrhagic fever] | Paramyxovirus [Humans: colds, respiratory infections, mumps]; Pneumovirus [Humans: pneumonia, common cold]; Morbillivirus [Humans: measles] |

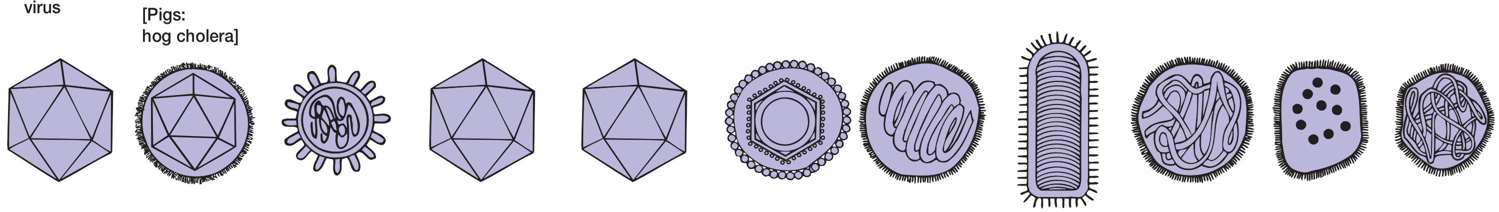


**Figure 18.2** **The Taxonomy of RNA Animal Viruses.**

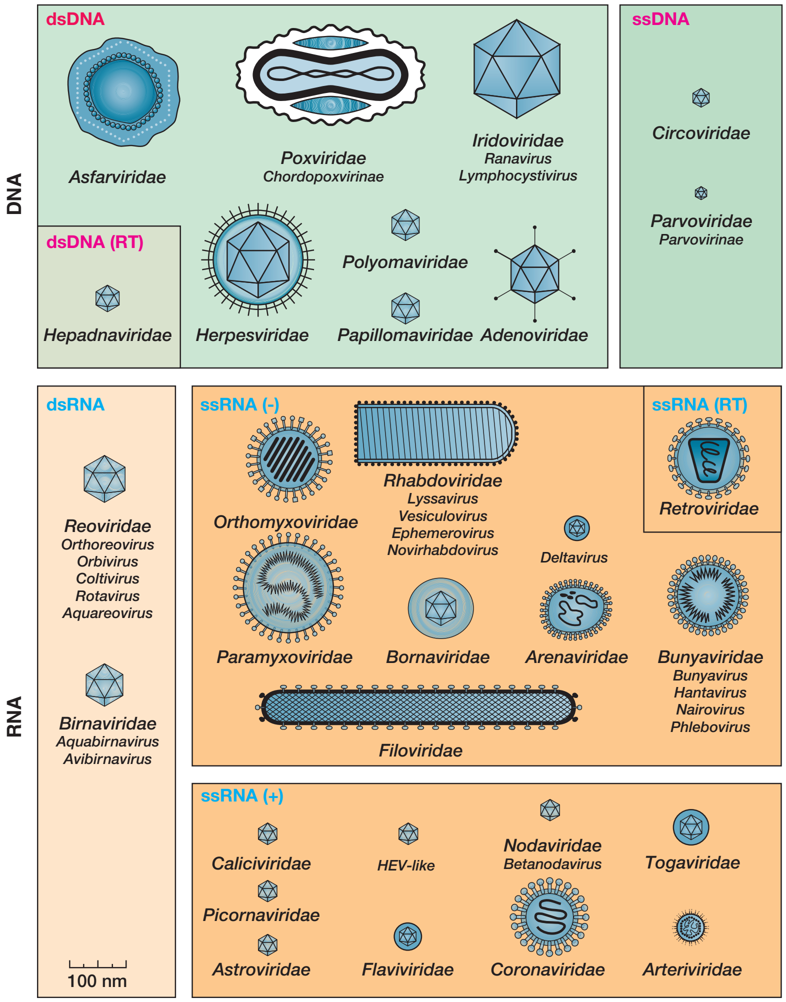



**Figure 18.3 A Diagrammatic Description of the Families and Genera of Viruses That Infect Vertebrates.** RT stands for reverse transcriptase.

**Table 18.1** Examples of Host Cell Surface Proteins That Serve as Virus Receptors

| Virus | Cell Surface Protein |
|---|---|
| Adenovirus | Coxsackie adenovirus receptor (CAR) protein |
| Epstein-Barr virus | Receptor for the C3d complement protein on human B lymphocytes |
| Hepatitis A virus | Alpha 2-macroglobulin |
| Herpes simplex virus, type 1 | Fibroblast growth factor receptor; a member of the tumor necrosis factor/nerve growth factor receptor family |
| Human immunodeficiency virus | CD4 protein on T-helper cells, macrophages, and monocytes; CXCR-4 or the CCR5 receptor |
| Influenza A virus | Sialic acid–containing glycoprotein |
| Measles virus | CD46 complement regulator protein |
| Poliovirus | Immunoglobulin superfamily |
| Rabies virus | Acetylcholine receptor on neurons |
| Rhinovirus | Intercellular adhesion molecules (ICAMs) on the surface of respiratory epithelial cells |
| Reovirus, type 3 | β-adrenergic receptor |
| Rotavirus | Acetylated sialic acid on glycoprotein |
| Vaccinia virus | Epidermal growth factor receptor |

Many host receptors are members of the immunoglobulin superfamily, a group of molecules that contain immunoglobulin domains (*see p. 734*). Most members of the immunoglobulin superfamily are surface proteins that are involved in the immune response and cell-cell interactions. Examples are the HIV CD4 receptor, the rhinovirus receptor, and the poliovirus ICAM (*i*ntercellular *a*dhesion *m*olecule) receptor. In some cases, it appears that two or more host cell receptors are involved in attachment. Herpes simplex virus interacts with a glycosaminoglycan and a member of the tumor necrosis factor/nerve growth factor receptor family. HIV uses CD4 and CXCR-4 (fusin) or the CCR5 (CC-CKR-5) receptor, both chemokine receptors.

The surface site on the virus can consist simply of a capsid structural protein or an array of such proteins. In some viruses—for example, the poliovirus and rhinoviruses—the binding site is at the bottom of a surface depression or valley. The site can bind to host cell surface projections but cannot be reached by host antibodies. Envelope glycoproteins also can be involved in the adsorption and penetration of enveloped viruses. For example, the herpes simplex virus has two envelope glycoproteins that are required for attachment, and at least four glycoproteins participate in penetration. In other cases the virus attaches to the host cell through special projections such as the fibers extending from the corners of adenovirus icosahedrons (*see figure 16.12*h) or the spikes of enveloped viruses. As mentioned in chapter 16, the influenza virus has two kinds of spikes: hemagglutinin and neuraminidase (*see figure 16.17*a,b). The hemagglutinin spikes appear to be involved in attachment to the host cell receptor site and recognize sialic acid (N-acetylneuraminic acid). Human virus diseases (chapter 38)

## Penetration and Uncoating

Viruses penetrate the plasma membrane and enter a host cell shortly after adsorption. Virus uncoating, the removal of the capsid and release of viral nucleic acid, occurs during or shortly after penetration. Both events will be reviewed together. The mechanisms of penetration and uncoating must vary with the type of virus because viruses differ so greatly in structure and mode of reproduction. For example, enveloped viruses may enter cells in a different way than naked or unenveloped virions. Furthermore, some viruses inject only their nucleic acid, whereas others must ensure that a virus-associated RNA or DNA polymerase also enters the host cell along with the virus genome. The entire process from adsorption to final uncoating may take from minutes to several hours.

The detailed mechanisms of penetration and uncoating are still unclear; it is possible that three different modes of entry may be employed (**figure 18.4**).

1. At least some naked viruses such as the poliovirus undergo a major change in capsid structure on adsorption to the plasma membrane, so that only their nucleic acids are released in the cytoplasm.
2. The envelope of paramyxoviruses, and possibly some other enveloped viruses, seems to fuse directly with the host cell plasma membrane. Fusion may involve special envelope fusion glycoproteins that bind to plasma membrane proteins. Then two things happen: membrane lipids rearrange and the adjacent halves of the contacting membranes merge, and a proteinaceous fusion pore forms. Finally the nucleocapsid enters the host cell cytoplasmic matrix, where uncoating is completed. A virus polymerase, associated with the nucleocapsid, begins transcribing the virus RNA while it is still within the capsid.
3. Most enveloped viruses may enter cells by a third route: through engulfment by receptor-mediated endocytosis to form **coated vesicles.** The virions attach to coated pits, specialized membrane regions coated on the cytoplasmic side with the protein clathrin. The pits then pinch off to form coated vesicles filled with viruses, and these fuse with lysosomes after the clathrin has been removed. Lysosomal enzymes may aid in virus uncoating and low endosomal pHs often trigger the uncoating process. In at least some instances, the virus envelope fuses with the lysosomal membrane, and the nucleocapsid is released into the cytoplasmic matrix (the capsid proteins may have been partially removed by lysosomal enzymes). Once in the cytoplasm, viral nucleic acid may be released from the capsid upon completion of uncoating or may function while still attached to capsid components.

## Replication and Transcription in DNA Viruses

The early part of the synthetic phase, governed by the **early genes,** is devoted to taking over the host cell and to the synthesis of viral DNA and RNA. Some virulent animal viruses inhibit host cell DNA, RNA, and protein synthesis, though cellular DNA is not usually degraded. In contrast, nonvirulent viruses may actually stimulate the synthesis of host macromolecules. DNA replication usually occurs in the host cell nucleus; poxviruses are exceptions since their genomes are replicated in the cytoplasm. Messenger RNA—at least early mRNA—is transcribed from DNA by host enzymes, except for poxvirus early mRNA, which is synthesized by a viral polymerase. Some examples of DNA virus reproduction will help illustrate these generalizations.

Parvoviruses, with a genome composed of one small, single-stranded DNA molecule about 4,800 bases in size, are the simplest of the DNA viruses (*see figure 16.12*a). The genome is so small that it directs the synthesis of only three polypeptides, all capsid components. Even so, the virus must resort to the use of overlapping genes to fit three genes into such a small molecule. That is, the base sequences that code for the three polypeptide chains overlap each other and are read with different reading frames (*see section 11.5*). Since the genome does not code for any enzymes, the virus must use host cell enzymes for all biosynthetic processes. Thus viral DNA can only be replicated in the nucleus during the S period of the cell cycle, when the cell replicates its own DNA. The genetic code and gene structure (pp. 240–44)

Herpesviruses are a large group of icosahedral, enveloped, double-stranded DNA viruses responsible for many important

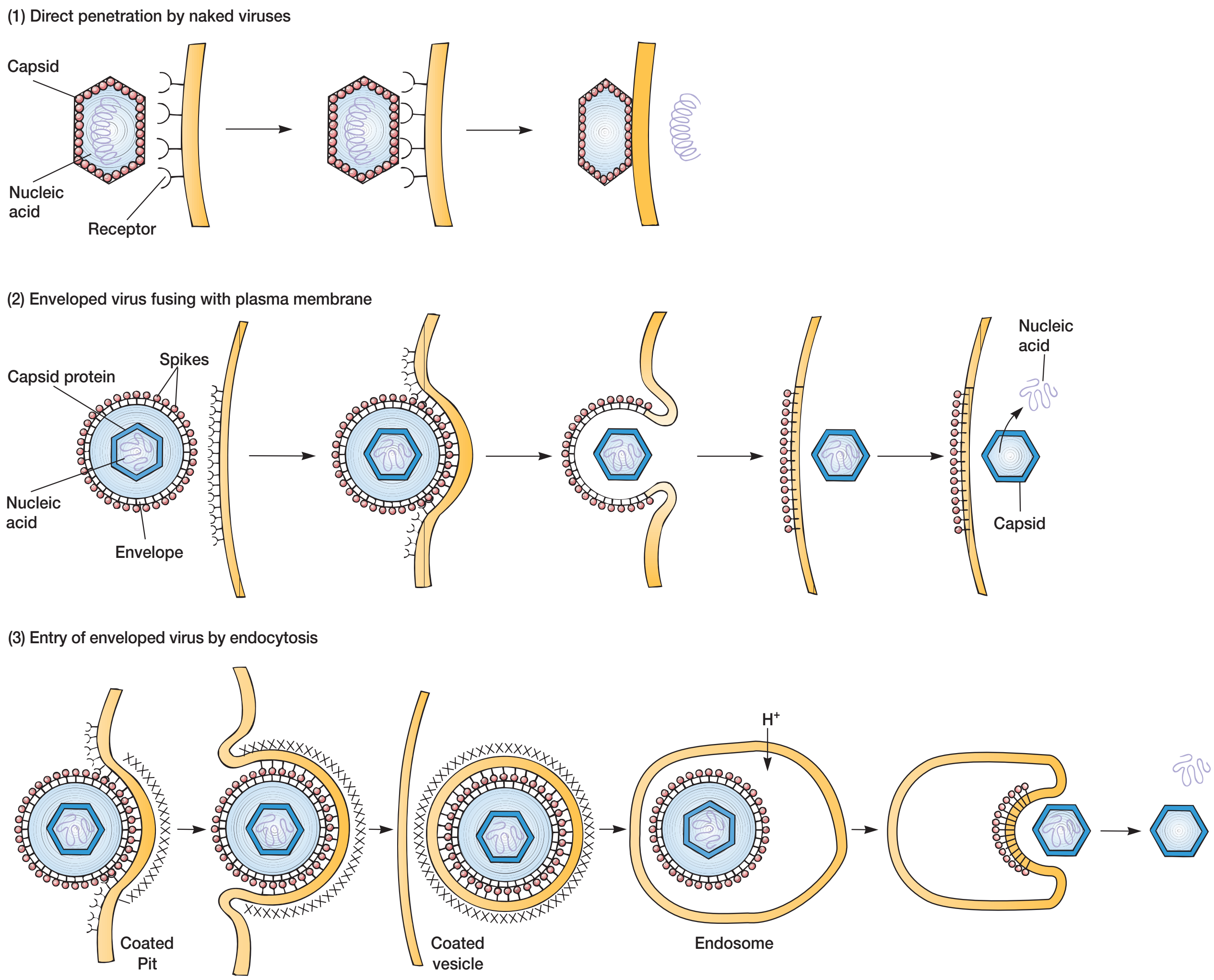



**Figure 18.4 Animal Virus Entry.** Mechanisms of animal virus attachment and entry into host cells. See text for description of the three entry modes.

human and animal diseases (*see figure 16.17e*). The genome is a linear piece of DNA about 160,000 base pairs in size and contains at least 50 to 100 genes. Immediately upon uncoating, the DNA is transcribed by host RNA polymerase to form mRNAs directing the synthesis of several early proteins, principally regulatory proteins and the enzymes required for virus DNA replication (**figure 18.5,** steps 1 and 2). The DNA circularizes and replication with a virus-specific DNA polymerase begins in the cell nucleus within 4 hours after infection (step 3). Host DNA synthesis gradually slows during a lethal virus infection (not all herpes infections result in immediate cell death). Herpesviruses and disease (pp. 871–72, 884–87)

Poxviruses such as the vaccinia virus are the largest viruses known and are morphologically complex (*see figure 16.18*). Their double-stranded DNA possesses over 200 genes. The virus enters through receptor-mediated endocytosis in coated vesicles; the central core escapes from the lysosome and enters the cytoplasmic matrix. The core contains both DNA and a DNA-dependent RNA polymerase that synthesizes early mRNAs, one of which directs the production of an enzyme that completes virus uncoating. DNA polymerase and other enzymes needed for DNA replication are also synthesized early in the reproductive cycle, and replication begins about 1.5 hours after infection. About the time DNA replication commences, late mRNA transcription is initiated. Many late proteins are structural proteins used in capsid construction. The complete reproductive cycle in poxviruses takes about 24 hours. Poxvirus diseases (p. 876)

The hepadnaviruses such as hepatitis B virus are quite different from other DNA viruses with respect to genome replication. They have circular dsDNA genomes but replicate their DNA using the enzyme reverse transcriptase (p. 407). After in-

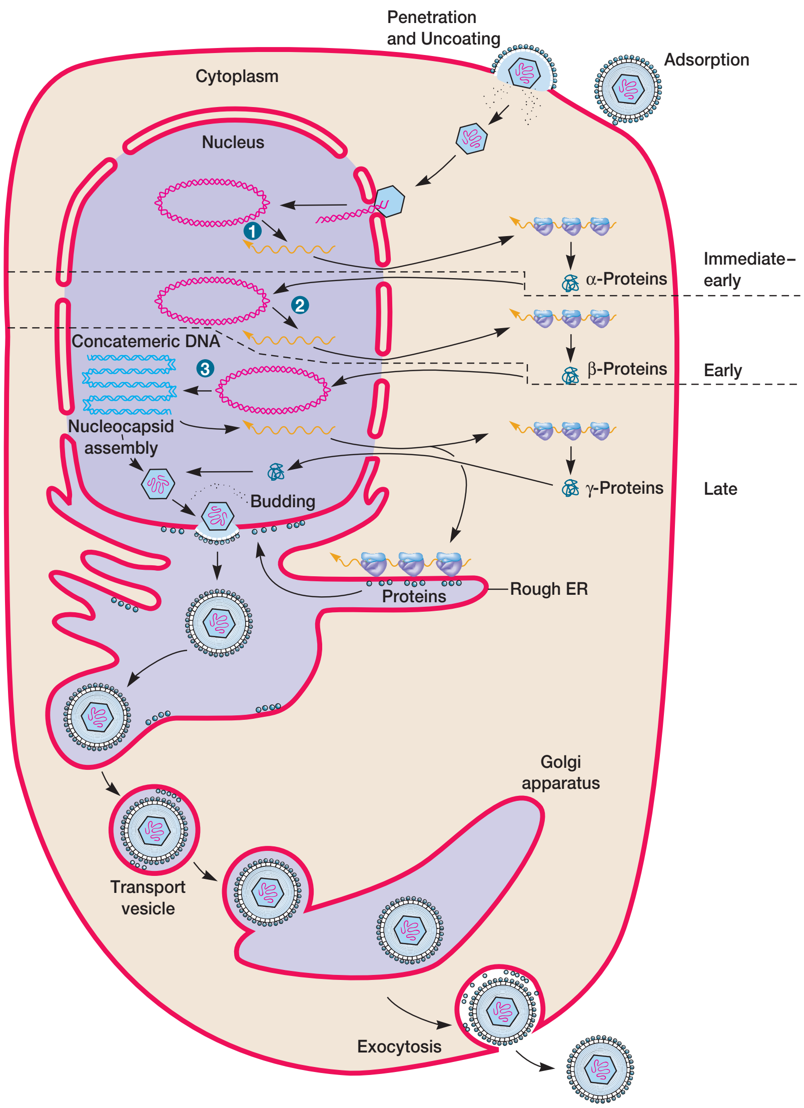



**Figure 18.5 Generalized Life Cycle for Herpes Simplex Virus Type I.** Only one possible pathway for release of the virions is shown. See text for details.

fecting a cell, their DNA is released in the nucleus. Transcription occurs in the nucleus using host RNA polymerase and yields several mRNAs, including a large 3.4 kilobase RNA known as the pregenome. The RNAs move to the cytoplasm and are translated to produce virus proteins such as core proteins and a polymerase having three activities (DNA polymerase, reverse transcriptase, RNase H). Then the RNA pregenome associates with DNA polymerase and core protein to form an immature core particle. Reverse transcriptase subsequently transcribes the RNA using a protein primer to form a −DNA copy of the pregenome +RNA. After almost all the pregenome RNA has been degraded by RNase H, the remaining RNA fragment serves as a primer for DNA polymerase to copy the minus DNA and form a dsDNA genome. Finally the nucleocapsid is completed. Hepatitis B virus and disease (pp. 889–90)

## Replication and Transcription in RNA Viruses

The RNA viruses are much more diverse in their reproductive strategies than are the DNA viruses. Most RNA viruses can be placed in one of four general groups based on their modes of replication and transcription, and their relationship to the host cell genome. **Figure 18.6** summarizes the reproductive cycles characteristic of these groups. Transcription patterns are discussed first, and then mechanisms of RNA replication.

Transcription in RNA viruses other than the retroviruses (retroviruses are considered shortly) varies with the nature of the virus genome. The picornaviruses such as poliovirus are the best studied positive strand ssRNA viruses. They use their RNA genome as a giant mRNA, and host ribosomes synthesize an enormous peptide that is then cleaved or processed by both host and viral encoded enzymes

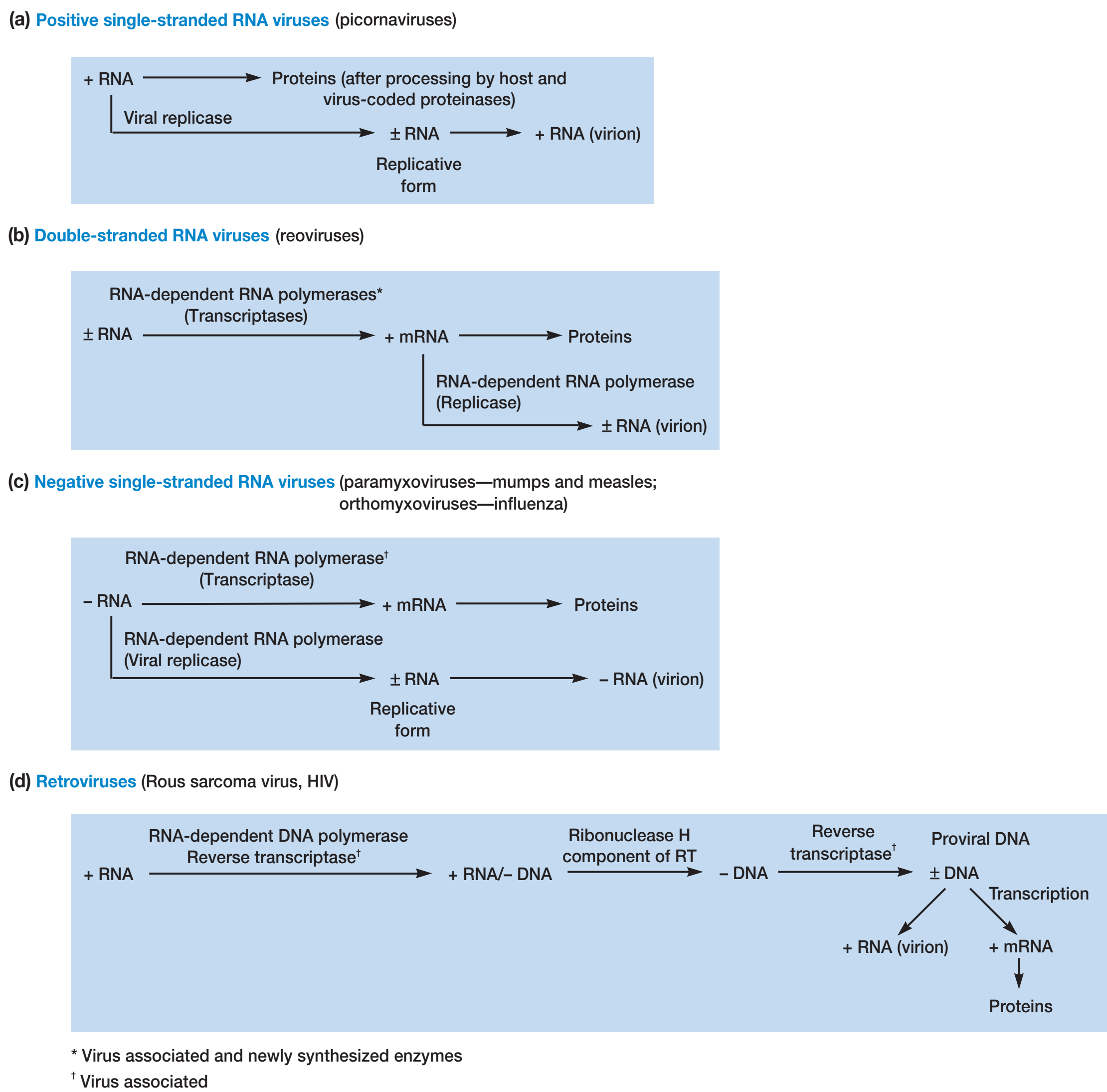



**Figure 18.6 RNA Animal Virus Reproductive Strategies.** Examples of viruses using the four strategies are given in parentheses.

to form the proper polypeptides (figure 18.6*a*). In contrast, because their genome is complementary to the mRNA base sequence, negative strand ssRNA viruses (e.g., the orthomyxoviruses and paramyxoviruses) must employ a virus-associated RNA-dependent RNA polymerase or **transcriptase** to synthesize mRNA (figures 18.6*c* and figure 18.7, step 1). The dsRNA reoviruses carry a virus-associated RNA-dependent RNA polymerase that copies the negative strand of their genome to generate mRNA. Later a new, virus-encoded polymerase continues transcription.

The nature of RNA replication also varies logically with the type of virus genetic material. Viral RNA is replicated in the host cell cytoplasmic matrix. Single-stranded RNA viruses, except retroviruses, use a viral **replicase** that converts the ssRNA into a double-stranded RNA called the **replicative form** (figure 18.6*a,c*). The appropriate strand of this intermediate then directs the synthesis of new viral RNA genomes (**figure 18.7,** step 2). Another way of looking at this process is that replication is governed by the principle of complementary. The parental genome strand directs the synthesis of a complementary strand, which then serves as a template for the synthesis of new progeny virus genomes. For example, picornaviruses have a +RNA genome and must make an intermediate −RNA complementary copy in order to produce new +RNA genomes. Reoviruses differ significantly from this pattern (figure 18.6*b*). The virion contains 10 to 13 different dsRNAs, each coding for an mRNA. Late in the reproductive cycle, a copy of each mRNA associates with the other

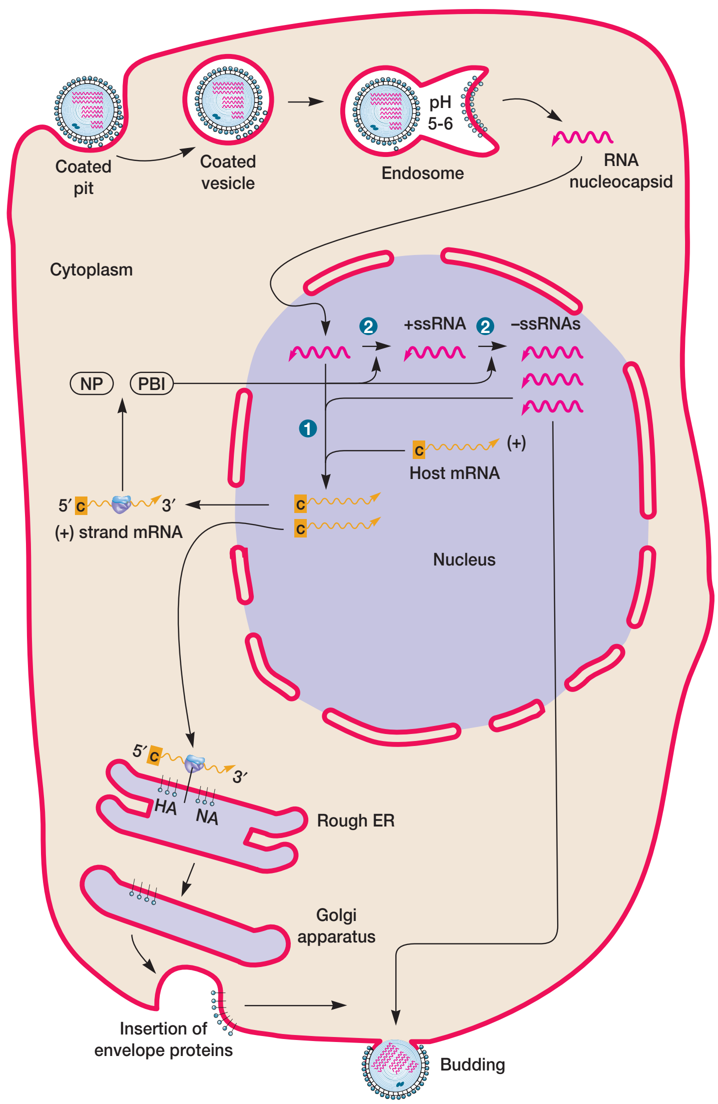



**Figure 18.7 Simplified Life Cycle of Influenza Virus.** Several steps have been eliminated for simplicity and clarity. Transcription (step 1) requires a process called cap snatching in which host RNAs are cleaved about 10 to 13 nucleotides from the 5′ end and the capped fragments are used as primers for the synthesis of viral mRNA. Abbreviations: PB1, RNA polymerase; NP, nucleocapsid protein; HA, hemagglutinin; and NA, neuraminidase.

mRNAs and special proteins to form a large complex. The RNAs in this complex are then copied by the viral replicase to form a double-stranded genome that is incorporated into a new virion. Reoviruses, orthomyxoviruses, and paramyxoviruses normally use a single polymerase for replication and transcription. Its mode of action depends on associated proteins and other factors.

**Retroviruses** such as the human immunodeficiency virus (*see figure 16.17*d) possess ssRNA genomes but differ from other RNA viruses in that they synthesize mRNA and replicate their genome by means of DNA intermediates. The virus has an RNA-dependent DNA polymerase or **reverse transcriptase (RT)** that copies the +RNA genome to form a −DNA copy (chapter opening figure and figure 18.6*d*; *see also figure 38.9*). Interestingly, transfer RNA is carried by the virus and serves as the primer required for nucleic acid synthesis (*see section 11.3*). The transformation of RNA into DNA takes place in two steps. First, reverse transcriptase copies the +RNA to form a RNA-DNA hybrid. Then the **ribonuclease H** component of reverse transcriptase degrades the +RNA strand to leave −DNA. After synthesizing –DNA, the reverse transcriptase copies this strand to produce a double-stranded DNA called **proviral DNA,** which can direct the synthesis of mRNA and new +RNA virion genome copies. Notice that during this process genetic information is transferred from RNA to DNA rather than in the normal direction.

The reproduction of retroviruses is remarkable in other ways as well. After proviral DNA has been manufactured, it is converted to a circular form and incorporated or integrated into the host cell chromosome. Virus products are only formed after integration. Sometimes these integrated viruses can change host cells into tumor cells. Biology of the HIV virus and AIDS (pp. 878–84)

**Table 18.2** Intracellular Sites of Animal Virus Reproduction

| Virus | Nucleic Acid Replication | Capsid Assembly | Membrane Used in Budding |
|---|---|---|---|
| **DNA Viruses** | | | |
| Adenoviruses | Nucleus | Nucleus | |
| Hepadnaviruses | Cytoplasm | Cytoplasm | Endoplasmic reticulum |
| Herpesviruses | Nucleus | At nuclear membrane | Nucleus |
| Papillomaviruses | Nucleus | Nucleus | |
| Parvoviruses | Nucleus | Nucleus | |
| Polyomaviruses | Nucleus | Nucleus | |
| Poxviruses | Cytoplasm | Cytoplasm | |
| **RNA Viruses** | | | |
| Coronaviruses | Cytoplasm | Cytoplasm | Golgi apparatus and endoplasmic reticulum |
| Orthomyxoviruses | Nucleus | Cytoplasm | Plasma membrane |
| Paramyxoviruses | Cytoplasm | Cytoplasm | Plasma membrane |
| Picornaviruses | Cytoplasm | Cytoplasm | |
| Reoviruses | Cytoplasm | Cytoplasm | |
| Retroviruses | Cytoplasm and nucleus | At plasma membrane | Plasma membrane |
| Rhabdoviruses | Cytoplasm | Cytoplasm | Plasma membrane, intracytoplasmic membranes |
| Togaviruses | Cytoplasm | Cytoplasm | Plasma membrane, intracytoplasmic membranes |

## Synthesis and Assembly of Virus Capsids

Some **late genes** direct the synthesis of capsid proteins, and these spontaneously self-assemble to form the capsid just as in bacteriophage morphogenesis. Recently the self-assembly process has been dramatically demonstrated. The addition of poliovirus RNA to an extract prepared from uninfected human cells (HeLa cells) results in the formation of new, infectious poliovirus virions. It appears that during icosahedral virus assembly empty **procapsids** are first formed; then the nucleic acid is inserted in some unknown way. The site of morphogenesis varies with the virus (**table 18.2**). Large paracrystalline clusters of complete virions or procapsids are often seen at the site of virus maturation (**figure 18.8**). The assembly of enveloped virus capsids is generally similar to that of naked virions, except for poxviruses. These are assembled in the cytoplasm by a lengthy, complex process that begins with the enclosure of a portion of the cytoplasmic matrix through construction of a new membrane. Then newly synthesized DNA condenses, passes through the membrane, and moves to the center of the immature virus. Nucleoid and elliptical body construction takes place within the membrane.

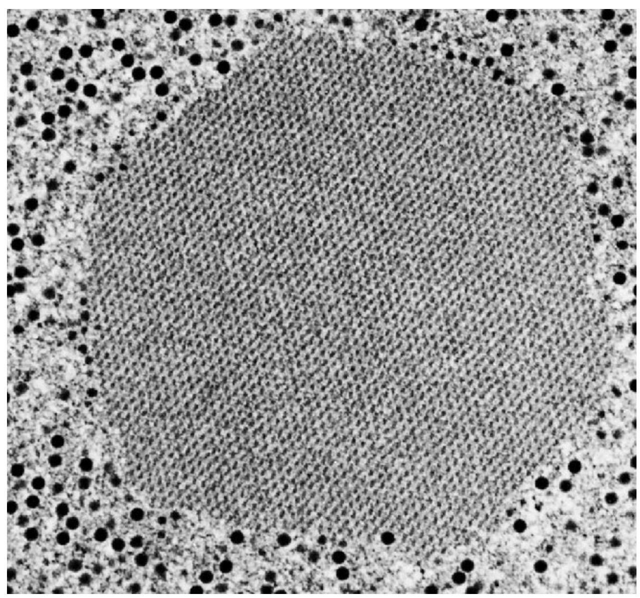


**Figure 18.8 Paracrystalline Clusters.** A crystalline array of adenoviruses within the cell nucleus (×35,000).

## Virion Release

Mechanisms of virion release differ between naked and enveloped viruses. Naked virions appear to be released most often by host cell lysis. In contrast, the formation of envelopes and the release of enveloped viruses are usually concurrent processes, and the host cell may continue virion release for some time. First, virus-encoded proteins are incorporated into the plasma membrane. Then the nucleocapsid is simultaneously released and the envelope formed by membrane budding (**figures 18.9** and **18.10**). In several virus families, a special M protein or matrix protein attaches to the plasma membrane and aids in budding. Although most envelopes arise from an altered plasma membrane, in herpesviruses, budding and envelope formation usually involves the nuclear envelope (table 18.2). The endoplasmic reticulum, Golgi apparatus, and other internal membranes also can be used to form envelopes.

Interestingly, it has been discovered that actin filaments can aid in virion release. Many viruses alter the actin microfilaments of the host cell cytoskeleton (*see p. 79*). For example, vaccinia virus appears to form long actin tails and use them to move intracellularly at up to 2.8 μm per minute. The actin filaments also propel vaccinia through the plasma membrane. In this way the virion escapes without destroying the host cell and infects adjacent cells.

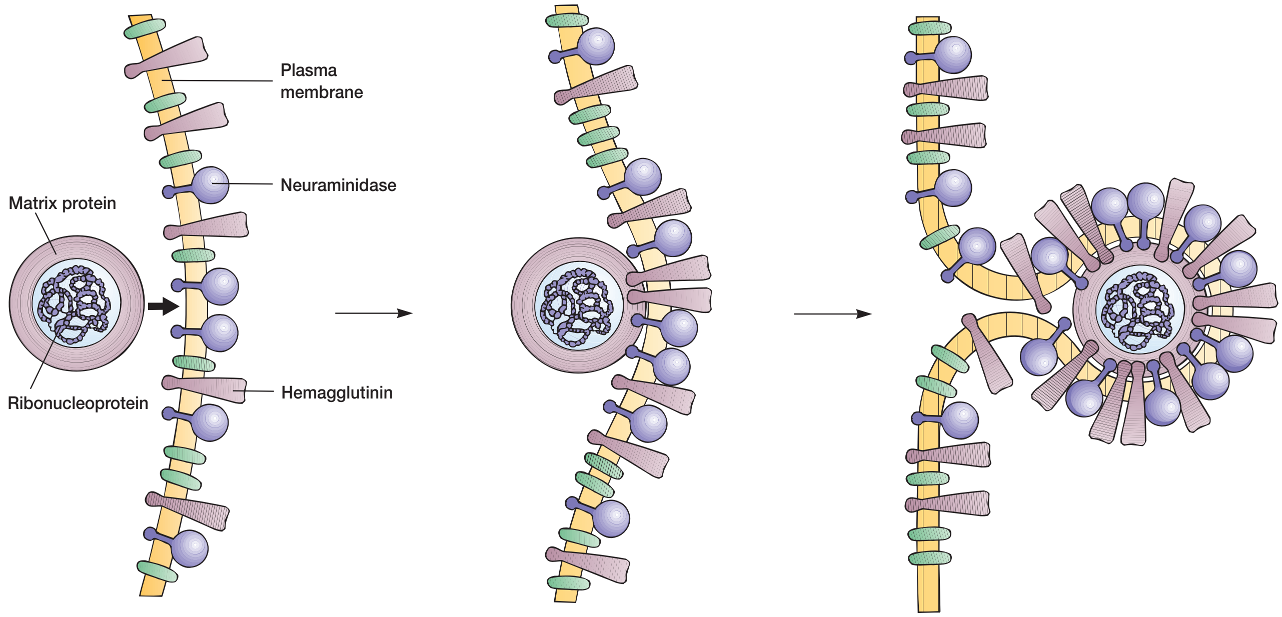
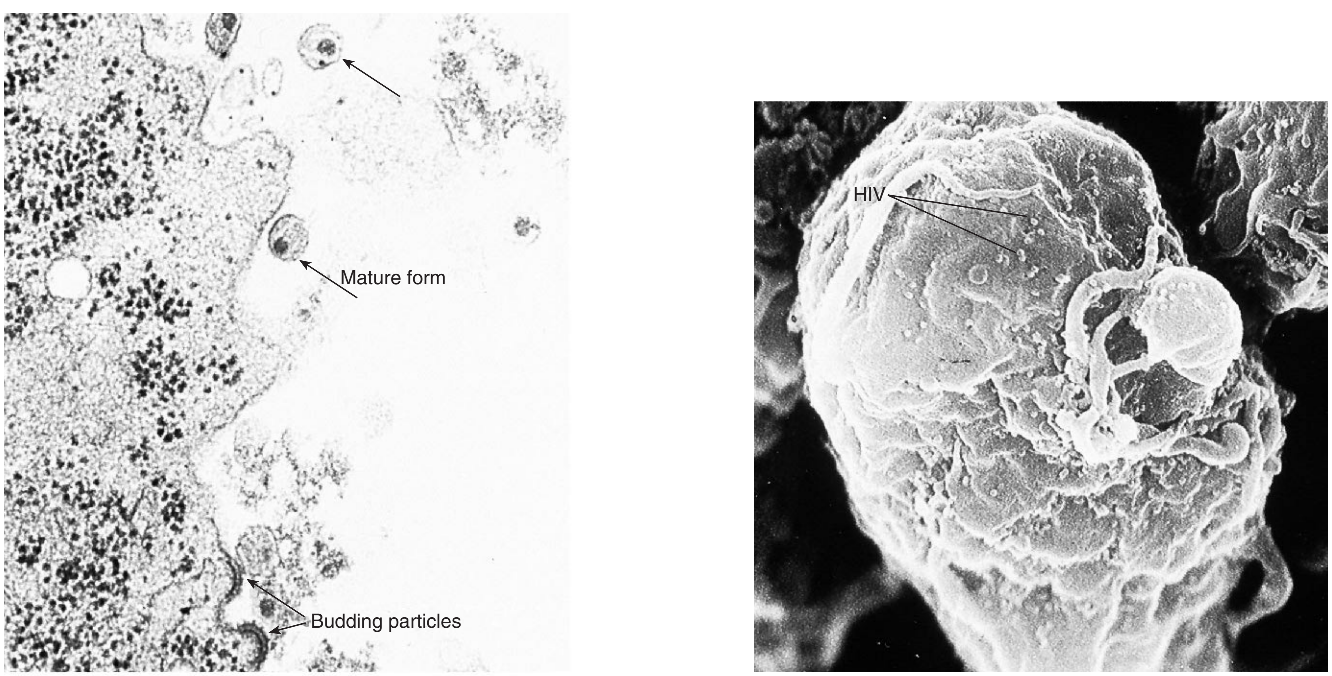



**Figure 18.9 Release of Influenza Virus by Plasma Membrane Budding.** First, viral envelope proteins (hemagglutinin and neuraminidase) are inserted into the host plasma membrane. Then the nucleocapsid approaches the inner surface of the membrane and binds to it. At the same time viral proteins collect at the site and host membrane proteins are excluded. Finally, the plasma membrane buds to simultaneously form the viral envelope and release the mature virion.




(a) (b)

**Figure 18.10 Human Immunodeficiency Virus (HIV) Release by Plasma Membrane Budding.** (**a**) A transmission electron micrograph of HIV particles beginning to bud, as well as some mature particles. (**b**) A scanning electron micrograph view of HIV particles budding from a lymphocyte.

1. Describe in detail each stage in animal virus reproduction and contrast it with the same stage in the bacteriophage life cycle (*see chapter 17*).
2. What probably plays the most important role in determining the tissue and host specificity of viruses?
3. Discuss how animal viruses can enter host cells.
4. Describe in general terms the replication and transcription processes that take place during animal virus reproduction.
5. Summarize the ways in which picornaviruses, –ssRNA viruses, and reoviruses carry out transcription and replication.
6. Outline in detail the life cycle of the retroviruses. What is proviral DNA?
7. In what ways are animal viruses released from their host cells?
8. Define the following terms: uncoating, coated vesicles, overlapping genes, replicative form, reverse transcriptase, and late genes.

## 18.3 Cytocidal Infections and Cell Damage

An infection that results in cell death is a **cytocidal infection.** Animal viruses can harm their host cells in many ways; often this leads to cell death. As mentioned earlier (*see section 16.3*), microscopic or macroscopic degenerative changes or abnormalities in host cells and tissues are referred to as cytopathic effects (CPEs), and these often result from virus infections. Seven possible mechanisms of host cell damage are briefly described here.

1. Many viruses can inhibit host DNA, RNA, and protein synthesis. Cytocidal viruses (e.g., picornaviruses, herpesviruses, and adenoviruses) are particularly active in this regard. The mechanisms of inhibition are not yet clear.
2. Cell lysosomes may be damaged, resulting in the release of hydrolytic enzymes and cell destruction.
3. Virus infection can drastically alter plasma membranes through the insertion of virus-specific proteins so that the infected cells are attacked by the immune system. When infected by viruses such as herpesviruses and measles virus, as many as 50 to 100 cells may fuse into one abnormal, giant, multinucleate cell or syncytium called a polykaryocyte. The HIV virus, which causes AIDS (*see chapter 38*), appears to destroy $CD4^+$ T-helper cells at least partly through its effects on the plasma membrane.
4. High concentrations of proteins from several viruses (e.g., mumps virus and influenza virus) can have a direct toxic effect on cells and organisms.
5. Intracellular structures called **inclusion bodies** are formed during many virus infections. These may result from the clustering of subunits or virions within the nucleus or cytoplasm (e.g., the Negri bodies in rabies infections); they also may contain cell components such as ribosomes (arenavirus infections) or chromatin (herpesviruses). Regardless of their composition, these inclusion bodies can directly disrupt cell structure.
6. Chromosomal disruptions result from infections by herpesviruses and others.
7. Finally, the host cell may not be directly destroyed but transformed into a malignant cell. This will be discussed later in the chapter.

It should be emphasized that more than one of these mechanisms may be involved in a cytopathic effect.

## 18.4 Persistent, Latent, and Slow Virus Infections

Many virus infections (e.g., influenza) are **acute infections**—that is, they have a fairly rapid onset and last for a relatively short time. However, some viruses can establish **persistent infections** lasting many years. There are several kinds of persistent infections. In **chronic virus infections,** the virus is almost always detectable and clinical symptoms may be either mild or absent for long periods. Examples are hepatitis B virus (serum hepatitis) and human immunodeficiency virus. In **latent virus infections** the virus stops reproducing and remains dormant for a period before becoming active again. During latency, no symptoms, antibodies, or viruses are detectable. Examples are herpes simplex virus, varicella-zoster virus, cytomegalovirus, and Epstein-Barr virus. Herpes simplex type 1 virus often infects children and then becomes dormant within the nervous system ganglia; years later it can be activated to cause cold sores. The varicella-zoster virus causes chicken pox in children and then, after years of inactivity, may produce the skin disease shingles (initial adult infections result in chicken pox). These and other examples of such infections are discussed in more detail in chapter 38.

The causes of persistence and latency are probably multiple, although the precise mechanisms are still unclear. The virus genome may be integrated into the host genome. Viruses may become less antigenic and thus less susceptible to attack by the immune system. They may mutate to less virulent and slower reproducing forms. Sometimes a deletion mutation (*see section 11.7*) produces a ***d**efective **i**nterfering* **(DI) particle** that cannot reproduce but slows normal virus reproduction, thereby reducing host damage and establishing a chronic infection.

A small group of viruses causes extremely slowly developing infections, often called **slow virus diseases** or slow infections, in which symptoms may take years to emerge. Measles virus occasionally produces a slow infection. A child may have a normal case of measles, then 5 to 12 years later develop a degenerative brain disease called subacute sclerosing panencephalitis (SSPE). Lentiviruses such as the human immunodeficiency virus also cause slow diseases. Many slow viruses may not be normal viruses at all. Several neurological diseases of animals and humans develop slowly. The best studied of these diseases are scrapie, a disease of sheep and goats, the human diseases kuru, fatal familial insomnia, Creutzfeldt-Jakob disease, and the new variant Creutzfeldt-Jakob disease. All are thought to be caused by simple, nonviral agents called prions, which are discussed later in the text.

1. Outline the ways in which viruses can damage host cells during cytocidal infections.
2. Define the following: acute infection, persistent infection, chronic infection, latent virus infection, and slow virus disease.
3. Why might an infection be chronic or latent?

## 18.5 Viruses and Cancer

**Cancer** [Latin *cancer,* crab] is one of our most serious medical problems and the focus of an immense amount of interest and research. A **tumor** [Latin *tumere,* to swell] is a growth or lump of tissue resulting from **neoplasia** or abnormal new cell growth and reproduction due to a loss of regulation. Tumor cells have aberrant shapes and altered plasma membranes that contain distinctive tumor antigens. They may invade surrounding tissues to form unorganized cell masses. They often lose the specialized metabolic activities characteristic of differentiated tissue cells and rely greatly upon anaerobic metabolism. This reversion to a more primitive or less differentiated state is called **anaplasia.**

There are two major types of tumors with respect to overall form or growth pattern. If the tumor cells remain in place to form a compact mass, the tumor is benign. In contrast, malignant or cancerous tumor cells can actively spread throughout the body in a process known as **metastasis,** often by floating in the blood and establishing secondary tumors. Some cancers are not solid, but cell suspensions. For example, leukemias are composed of malignant white blood cells that circulate throughout the body. Indeed, dozens of kinds of cancers arise from a variety of cell types and afflict all kinds of organisms.

As one might expect from the wide diversity of cancers, there are many causes of cancer, only some of which are directly related to viruses. Possibly as many as 30 to 60% of cancers may be related to diet. Many chemicals in our surroundings are carcinogenic and may cause cancer by inducing gene mutations or interfering with normal cell differentiation. The Ames test for carcinogens (pp. 253–54)

Carcinogenesis is a complex, multistep process. It can be initiated by a chemical, usually a mutagen, but a cancer does not appear to develop until at least one more triggering event (possibly exposure to another chemical carcinogen or a virus) takes place. Often, as is discussed later, cancer-causing genes, or **oncogenes,** are directly involved and may come from the cell itself or be contributed by a virus. Many of these oncogenes seem to be involved in the regulation of cell growth and differentiation; for example, some code for all or part of growth factors that regulate cell growth. It may be that various cancers arise through different combinations of causes. Not surprisingly the chances of developing cancer rise with age because an older person will have been exposed to carcinogens and other causative factors for a longer time. Immune surveillance and destruction of cancer cells (*see chapter 32*) also may be less effective in older people.

Although viruses are known to cause many animal cancers, it is very difficult to prove that this is the case with human cancers since indirect methods of study must be used and Koch's postulates can't be applied completely. One tries to find virus particles and components within tumor cells, using techniques such as electron microscopy, immunologic tests, and enzyme assays. Attempts are also made to isolate suspected cancer viruses by cultivation in tissue culture or other animals. Sometimes a good correlation between the presence of a virus and cancer can be detected.

At present, viruses have been implicated in the genesis of at least eight human cancers:

1. The Epstein-Barr virus (EBV) is one of the best-studied human cancer viruses. EBV is a herpesvirus and the cause of two cancers. Burkitt's lymphoma is a malignant tumor of the jaw and abdomen found in children of central and western Africa. EBV also causes nasopharyngeal carcinoma. Both the virus particles and the EBV genome have been found within tumor cells; Burkitt's lymphoma patients also have high blood levels of antibodies to EBV. Interestingly there is some reason to believe that a person also must have had malaria to develop Burkitt's lymphoma. Environmental factors must play a role, because EBV does not cause much cancer in the United States despite its prevalence. Possibly this is due to a low incidence of malaria in the United States.
2. Hepatitis B virus appears to be associated with one form of liver cancer (hepatocellular carcinoma) and can be integrated into the human genome.
3. Hepatitis C virus causes cirrhosis of the liver, which can lead to liver cancer.
4. Human herpesvirus 8 is associated with the development of Kaposi's sarcoma.
5. Some strains of human papillomaviruses have been linked to cervical cancer.
6. At least two retroviruses, the human T-cell lymphotropic virus I (HTLV-1) and HTLV-2, seem able to cause cancer, adult T-cell leukemia, and hairy-cell leukemia, respectively (*see figure 38.17*). Other retrovirus-associated cancers may well be discovered in the future.

It appears that viruses can cause cancer in several ways. They may bring oncogenes into a cell and insert them into its genome. Rous sarcoma virus (a retrovirus) carries an *src* gene that codes for tyrosine kinase. This enzyme is located mainly in the plasma membrane and phosphorylates the amino acid tyrosine in several cellular proteins. It is thought that this alters cell growth and behavior. Since the activity of many proteins is regulated by phosphorylation and several other oncogenes also code for protein kinases, many cancers may result at least partly from altered cell regulation due to changes in kinase activity. The human T-cell lymphotropic viruses, HTLV-1 and HTLV-2, seem to transform T cells (*see chapter 32*) by producing a regulatory protein that sometimes activates genes involved in cell division as well as stimulating virus reproduction. Some oncogenic viruses carry one or more very effective promoters or enhancers (*see section 11.5*). If these viruses integrate themselves next to a cellular oncogene, the promoter or enhancer will stimulate its transcription, leading to cancer. In this case the oncogene might be necessary for normal cell growth (possibly it codes for a regulatory protein) and only causes cancer when it functions too rapidly or at the wrong time. For example, some chicken retroviruses induce lymphomas when they are integrated next to the *c-myc* cellular oncogene, which codes for a protein that appears to be involved in the induction of either DNA or RNA synthesis. With the possible exception of HTLV-1, it is not yet known how the viruses associated with human cancers actually aid in cancer development.

1. What are the major characteristics of cancer?
2. How might viruses cause cancer? Are there other ways in which a malignancy might develop?
3. Define the following terms: tumor, neoplasia, anaplasia, metastasis, and oncogene.

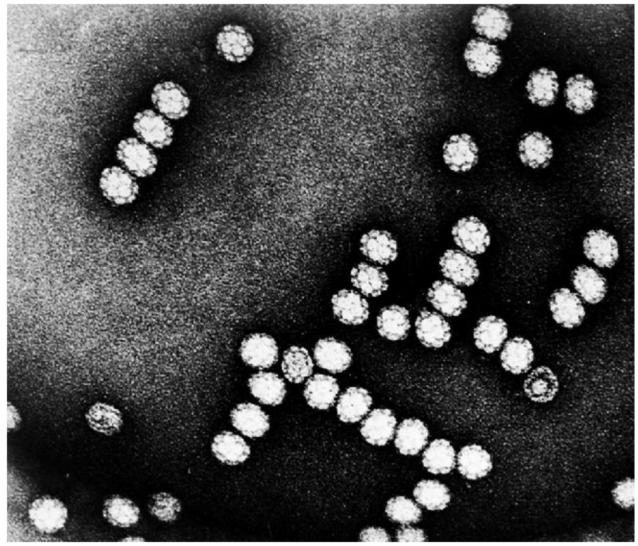


**Figure 18.11 Turnip Yellow Mosaic Virus (TYMV).** An RNA plant virus with icosahedral symmetry.

## 18.6 Plant Viruses

Although it has long been recognized that viruses can infect plants and cause a variety of diseases (*see figure 16.5*), plant viruses generally have not been as well studied as bacteriophages and animal viruses. This is mainly because they are often difficult to cultivate and purify. Some viruses, such as tobacco mosaic virus (TMV), can be grown in isolated protoplasts of plant cells just as phages and some animal viruses are cultivated in cell suspensions. However, many cannot grow in protoplast cultures and must be inoculated into whole plants or tissue preparations. Many plant viruses require insect vectors for transmission; some of these can be grown in monolayers of cell cultures derived from aphids, leafhoppers, or other insects.

### Virion Morphology

The essentials of capsid morphology are outlined in chapter 16 and apply to plant viruses since they do not differ significantly in construction from their animal virus and phage relatives. Many have either rigid or flexible helical capsids (tobacco mosaic virus, *see figure 16.11*). Others are icosahedral or have modified the icosahedral pattern with the addition of extra capsomers (turnip yellow mosaic virus, **figure 18.11**). Most capsids seem composed of one type of protein; no specialized attachment proteins have been detected. Almost all plant viruses are RNA viruses, either single stranded or double stranded (*see tables 16.1 and 16.2*). Caulimoviruses and geminiviruses with their DNA genomes are exceptions to this rule.

### Plant Virus Taxonomy

The shape, size, and nucleic acid content of many plant virus groups are summarized in **figure 18.12.** Like other types of viruses, they are classified according to properties such as nucleic acid type and strandedness, capsid symmetry and size, and envelope presence (*see table 16.2*).

### Plant Virus Reproduction

Since tobacco mosaic virus (TMV) has been studied the most extensively, its reproduction will be briefly described. The replication of virus RNA is an essential part of the reproduction process. Most plants contain RNA-dependent RNA polymerases, and it is possible that these normal constituents replicate the virus RNA. However, some plant virus genomes (e.g., turnip yellow virus and cowpea mosaic virus) appear to be copied by a virus-specific RNA replicase. Possibly TMV RNA is also replicated by a viral RNA polymerase, but the evidence is not clear on this matter. Four TMV-specific proteins, one of them the coat protein, are known to be made. Although TMV RNA is plus single-stranded RNA and could directly serve as mRNA, the production of messenger is complex. The coat protein mRNA has the same sequence as the 3′ end of the TMV genome and arises from it by some sort of intracellular processing.

After the coat protein and RNA genome have been synthesized, they spontaneously assemble into complete TMV virions in a highly organized process (**figure 18.13**). The protomers (*see section 16.5*) come together to form disks composed of two layers of protomers arranged in a helical spiral. Association of coat protein with TMV RNA begins at a special assembly initiation site close to the 3′ end of the genome. The helical capsid grows by the addition of protomers, probably as disks, to the end of the rod. As the rod lengthens, the RNA passes through a channel in its center and forms a loop at the growing end. In this way the RNA can easily fit as a spiral into the interior of the helical capsid.

Reproduction within the host depends on the virus's ability to spread throughout the plant. Viruses can move long distances through the plant vasculature; usually they travel in the phloem. The spread of plant viruses in nonvascular tissue is hindered by the presence of tough cell walls. Nevertheless, a virus such as TMV does spread slowly, about 1 mm/day or less. It moves from cell to cell through the plasmodesmata. These are slender cyto-

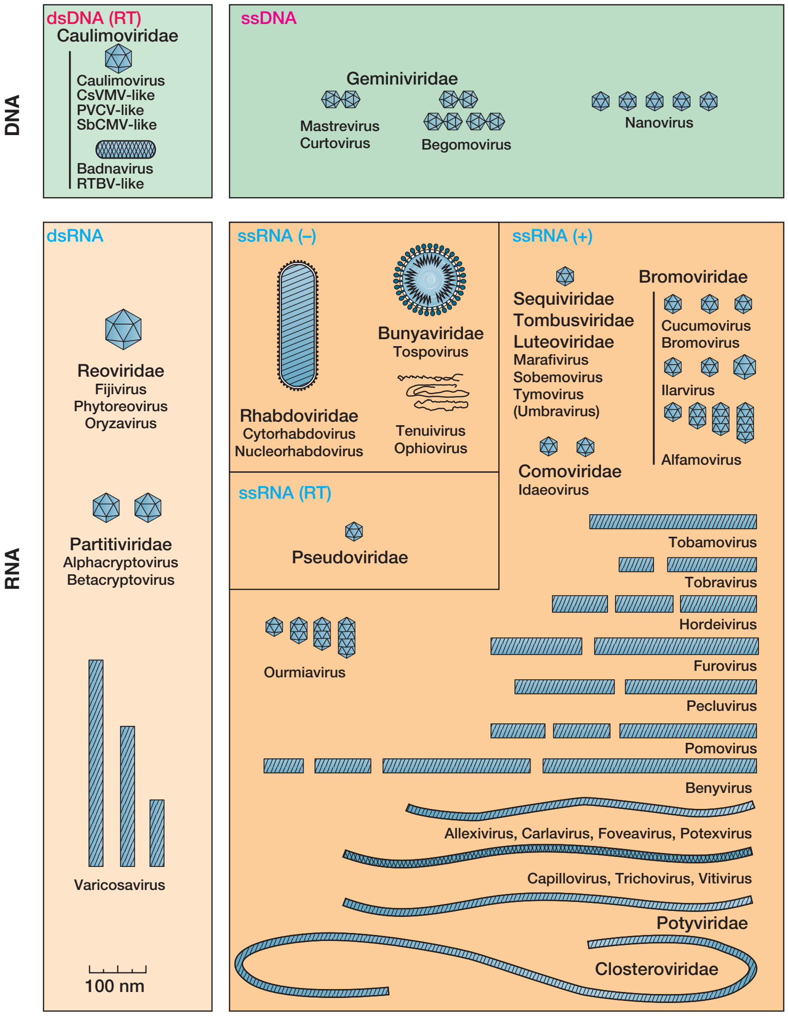



**Figure 18.12 A Diagrammatic Description of Families and Genera of Viruses That Infect Plants.** RT stands for reverse transcriptase.

plasmic strands extending through holes in adjacent cell walls that join plant cells by narrow bridges. Special viral "movement proteins" are required for movement between cells. The TMV movement protein accumulates in the plasmodesmata, but the way in which it promotes virus movement is not understood.

Several cytological changes can take place in TMV-infected cells. Plant virus infections often produce microscopically visible intracellular inclusions, usually composed of virion aggregates, and hexagonal crystals of almost pure TMV virions sometimes do develop in TMV-infected cells (**figure 18.14**). The host cell chloroplasts become abnormal and often degenerate, while new chloroplast synthesis is inhibited.

## Transmission of Plant Viruses

Since plant cells are protected by cell walls, plant viruses have a considerable obstacle to overcome when trying to establish themselves in a host. TMV and a few other viruses may be carried by the wind or animals and then enter when leaves are mechanically damaged. Some plant viruses are transmitted through contaminated seeds, tubers, or pollen. Soil nematodes can transmit viruses (e.g., the tobacco ringspot virus) while feeding on roots. Tobacco necrosis virus is transmitted by parasitic fungi. However, the most important agents of transmission are insects that feed on plants, particularly sucking insects such as aphids and leafhoppers.

Insects transmit viruses in several ways. They may simply pick up viruses on their mouth parts while feeding on an infected plant, then directly transfer the viruses to the next plant they visit. Viruses may be stored in an aphid's foregut; the aphid will infect plants when regurgitating while it is feeding. Several plant viruses—for example, the wound tumor virus—can multiply in leafhopper tissues before reaching the salivary glands and being inoculated into plants (i.e., it uses both insects and plants as hosts).

1. Why have plant viruses not been as well studied as animal and bacterial viruses?
2. Describe in molecular terms the way in which TMV is reproduced.
3. How are plant viruses transmitted between hosts?

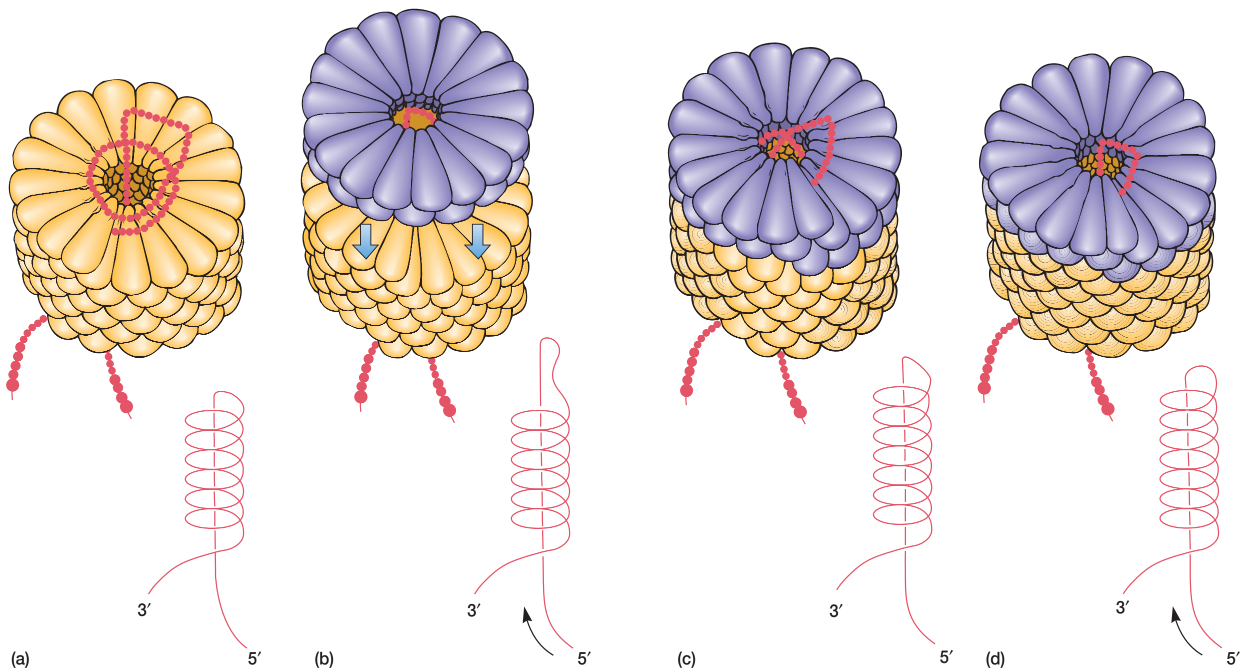
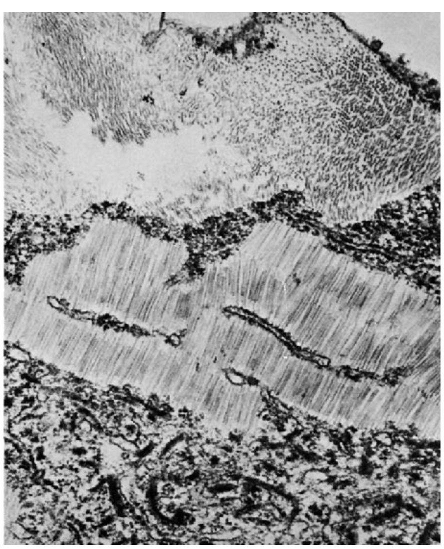
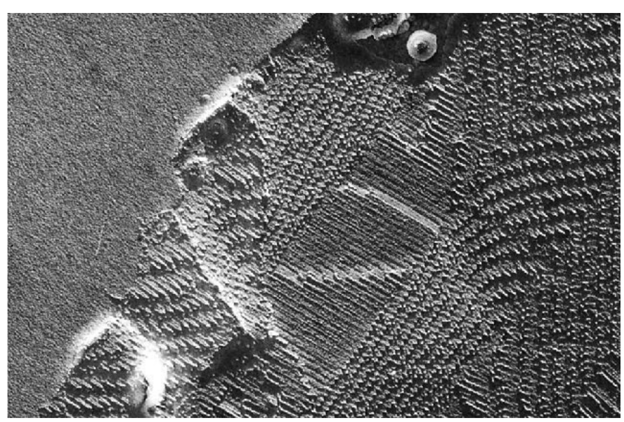



**Figure 18.13 TMV Assembly.** The elongation phase of tobacco mosaic virus nucleocapsid construction. The lengthening of the helical capsid through the addition of a protein disk to its end is shown in a sequence of four illustrations; line drawings depicting RNA behavior are included. The RNA genome inserts itself through the hole of an approaching disk and then binds to the groove in the disk as it locks into place at the end of the cylinder.



(a)



(b)

**Figure 18.14 Intracellular TMV.** (**a**) A crystalline mass of tobacco mosaic virions from a 10-day-old lesion in a *Chenopodium amaranticolor* leaf. (**b**) Freeze-fracture view of a crystalline mass of tobacco mosaic virions in an infected leaf cell. In both views the particles can be seen longitudinally and in cross section.

## 18.7 Viruses of Fungi and Algae

Most of the fungal viruses or mycoviruses isolated from higher fungi such as *Penicillium* and *Aspergillus* (*see chapter 25*) contain double-stranded RNA and have isometric capsids (that is, their capsids are roughly spherical polyhedra), which are approximately 25 to 50 nm in diameter. Many appear to be latent viruses. Some mycoviruses do induce disease symptoms in hosts such as the mushroom *Agaricus bisporus,* but cytopathic effects and toxic virus products have not yet been observed.

Much less is known about the viruses of lower fungi, and only a few have been examined in any detail. Both dsRNA and dsDNA genomes have been found; capsids usually are isometric or hexagonal and vary in size from 40 to over 200 nm. Unlike the situation in higher fungi, virus reproduction is accompanied by host cell destruction and lysis.

Viruses have been detected during ultrastructural studies of eucaryotic algae, but few have been isolated. Those that have been studied have dsDNA genomes and polyhedral capsids. One virus of the green alga *Uronema gigas* resembles many bacteriophages in having a tail.

1. Describe the major characteristics of the viruses that infect higher fungi, lower fungi, and algae. In what ways do they seem to differ from one another?

## 18.8 Insect Viruses

Members of at least seven virus families (*Baculoviridae, Iridoviridae, Poxviridae, Reoviridae, Parvoviridae, Picornaviridae,* and *Rhabdoviridae*) are known to infect insects and reproduce or even use them as the primary host (*see table 16.2*). Of these, probably the three most important are the *Baculoviridae, Reoviridae,* and *Iridoviridae.*

The *Iridoviridae* are icosahedral viruses with lipid in their capsids and a linear double-stranded DNA genome. They are responsible for the iridescent virus diseases of the crane fly and some beetles. The group's name comes from the observation that larvae of infected insects can have an iridescent coloration due to the presence of crystallized virions in their fat bodies.

Many insect virus infections are accompanied by the formation of inclusion bodies within the infected cells. Granulosis viruses form granular protein inclusions, usually in the cytoplasm. Nuclear polyhedrosis and cytoplasmic polyhedrosis virus infections produce polyhedral inclusion bodies in the nucleus or the cytoplasm of affected cells. Although all three types of viruses generate inclusion bodies, they belong to two distinctly different families. The cytoplasmic polyhedrosis viruses are reo-viruses; they are icosahedral with double shells and have double-stranded RNA genomes. Nuclear polyhedrosis viruses and granulosis viruses are baculoviruses—rod-shaped, enveloped viruses of helical symmetry and with double-stranded DNA.

The inclusion bodies, both polyhedral and granular, are protein in nature and enclose one or more virions (**figure 18.15**). Insect larvae are infected when they feed on leaves contaminated with inclusion bodies. Polyhedral bodies protect the virions against heat, low pH, and many chemicals; the viruses can remain viable in the soil for years. However, when exposed to alkaline insect gut contents, the inclusion bodies dissolve to liberate the virions, which then infect midgut cells. Some viruses remain in the midgut while others spread throughout the insect. Just as with bacterial and vertebrate viruses, insect viruses can persist in a latent state within the host for generations while producing no disease symptoms. A reappearance of the disease may be induced by chemicals, thermal shock, or even a change in the insect's diet.

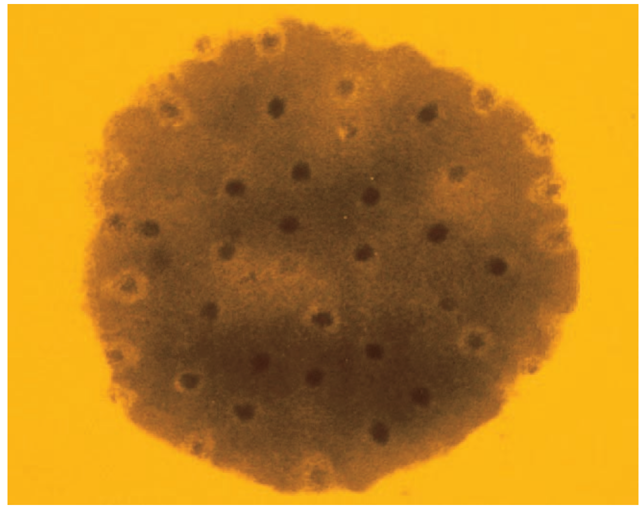


**Figure 18.15 Inclusion Bodies.** A section of a cytoplasmic polyhedron from a gypsy moth *(Lymantria dispar).* The occluded virus particles with dense cores are clearly visible (×50,000).

Much of the current interest in insect viruses arises from their promise as biological control agents for insect pests (*see chapter 42*). Many people hope that some of these viruses may partially replace the use of toxic chemical pesticides. Baculoviruses have received the most attention for at least three reasons. First, they attack only invertebrates and have considerable host specificity; this means that they should be fairly safe for nontarget organisms. Second, because they are encased in protective inclusion bodies, these viruses have a good shelf life and better viability when dispersed in the environment. Finally, they are well suited for commercial production since they often reach extremely high concentrations in larval tissue (as high as $10^{10}$ viruses per larva). The use of nuclear polyhedrosis viruses for the control of the cotton bollworm, Douglas fir tussock moth, gypsy moth, alfalfa looper, and European pine sawfly has either been approved by the Environmental Protection Agency or is being considered. The granulosis virus of the codling moth also is useful. Usually inclusion bodies are sprayed on foliage consumed by the target insects. More sensitive viruses are administered by releasing infected insects to spread the disease. As in the case of other pesticides, it is possible that resistance to these agents may develop in the future.

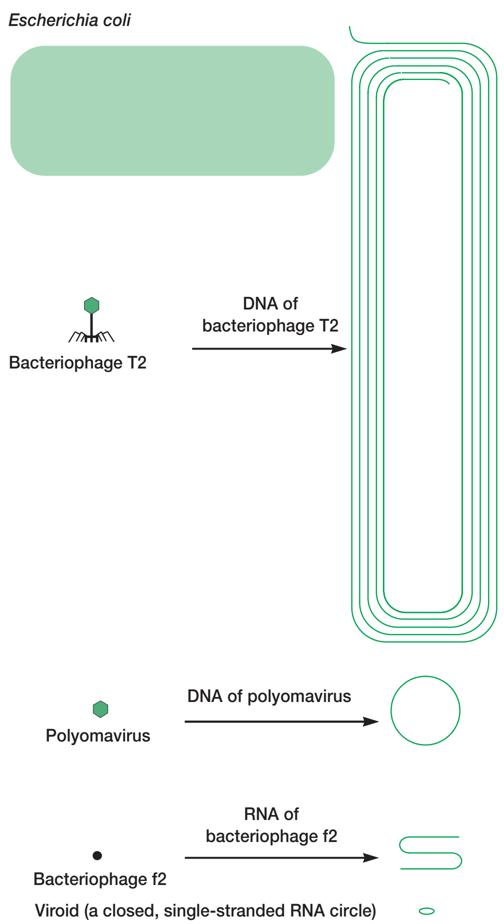



**Figure 18.16 Viroids, Viruses, and Bacteria.** A comparison of *Escherichia coli,* several viruses, and the potato spindle-tuber viroid with respect to size and the amount of nucleic acid possessed. (All dimensions are enlarged approximately ×40,000.)

1. Summarize the nature of granulosis, nuclear polyhedrosis, and cytoplasmic polyhedrosis viruses and the way in which they are transmitted by inclusion bodies.
2. What are baculoviruses and why are they so promising as biological control agents for insect pests?

## 18.9 Viroids and Prions

Although some viruses are exceedingly small and simple, even simpler infectious agents exist. Over 16 different plant diseases—for example, potato spindle-tuber disease, exocortis disease of citrus trees, and chrysanthemum stunt disease—are caused by a group of infectious agents called **viroids.** These are circular, single-stranded RNAs, about 250 to 370 nucleotides long (**figures 18.16** and **18.17**), that can be transmitted between plants through mechanical means or by way of pollen and ovules and are replicated in their hosts. The single-stranded RNA normally exists as a closed circle collapsed into a rodlike shape by intrastrand base pairing. Viroids are found principally in the nucleolus of infected cells; between 200 and 10,000 copies may be present. They do not act as mRNAs to direct protein synthesis, and it is not yet known how they cause disease symptoms. A plant may be infected without showing symptoms—that is, it may have a latent infection. The same viroid, when in another species, might well cause a severe disease. Although viroids could be replicated by an RNA-dependent RNA polymerase, they appear to be synthesized from RNA templates by a host RNA polymerase that mistakes them for a piece of DNA. A rolling-circle type mechanism seems to be involved (*see p. 236*).

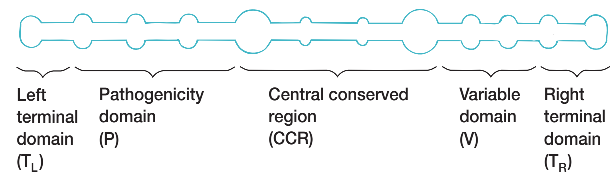



**Figure 18.17 Viroid Structure.** This schematic diagram shows the general organization of a viroid. The closed single-stranded RNA circle has extensive intrastrand base pairing and interspersed unpaired loops. Viroids have five domains. Most changes in viroid pathogenicity seem to arise from variations in the P and $T_L$ domains.

The potato spindle-tuber disease agent (PSTV) has been most intensely studied. Its RNA is about 130,000 daltons or 359 nucleotides in size, much smaller than any virus genome. Several PSTV strains have been isolated, ranging in virulence from ones that cause only mild symptoms to lethal varieties. All variations in pathogenicity are due to a few nucleotide changes in two short regions on the viroid. It is believed that these sequence changes alter the shape of the rod and thus its ability to cause disease.

There is evidence that an infectious agent different from both viruses and viroids can cause disease in livestock and humans. The agent has been called a **prion** (for proteinaceous infectious particle). The best studied of these prions causes a degenerative disorder of the central nervous system in sheep and goats; this disorder is named scrapie. Afflicted animals lose coordination of their movements, tend to scrape or rub their skin, and eventually cannot walk. No nucleic acid has yet been detected in the agent. It seems to be a 33 to 35 kDa hydrophobic membrane protein, often called PrP (for **pr**ion **p**rotein). The *PrP* gene is present in many normal vertebrates and invertebrates, and the prion protein is bound to the surface of neurons. Presumably an altered PrP is at least partly responsible for the disease.

Despite the isolation of PrP, the mechanism of prion diseases continues to stir controversy. Most researchers are convinced that prion diseases are transmitted by the PrP alone. They believe that the infective pathogen is an abnormal PrP ($PrP^{Sc}$; Sc, scrapie-associated), a protein that has either been changed in conformation or chemically modified. When $PrP^{Sc}$ enters a normal brain, it might bind to the normal cellular PrP ($PrP^{C}$) and induce it to fold into the abnormal conformation. The newly produced $PrP^{Sc}$ molecule could then convert more normal $PrP^C$ proteins to the $PrP^{Sc}$ form. Alternatively, $PrP^{Sc}$ could activate enzymes that modify PrP structure. Prions with different amino acid sequences and conformations convert $Prp^C$ molecules to $PrP^{Sc}$ in other hosts.

More support for the "protein-only" hypothesis has been supplied by studies on the yeast protein Sup35, which aids in the termination of protein synthesis in *Saccharomyces cerevisiae.* Because Sup35 acts much like mammalian prions, it has been called a yeast prion. Sup35 exists in an inactive form in [$PSI^+$] cells, and the phenotype can be passed on to daughter cells. The inactive, insoluble form of Sup35 aggregates and translation does not terminate properly. There is evidence that the [$PSI^+$] phenotype proliferates in yeast when the inactive prion form of Sup35 interacts with normal, soluble Sup35 and induces a self-propagating conformational change of active Sup35 to the inactive form.

A minority think that the "protein-only" hypothesis is inadequate. They are concerned about the existence of prion strains, which they believe are genetic, and about the problem of how genetic information can be transmitted between hosts by a protein. Doubters note that proteins have never been known to carry genetic information. It could be that prions somehow either directly aid an infectious agent such as an unknown virus or increase susceptibility to the agent. Possibly the real agent is a tiny nucleic acid that is coated with PrP and interacts with host cells to cause disease. This hypothesis is consistent with the finding that many strains of the scrapie agent have been isolated. There is also some evidence that a strain can change or mutate. Supporters of the "protein-only" hypothesis reply that strain characteristics are simply due to structural differences in the PrP molecule.

As mentioned earlier, some slow virus diseases may be due to prions or virinos. This is particularly true of certain neurological diseases of humans and animals. Bovine spongiform encephalopathy (BSE or "mad cow disease"), kuru, fatal familial insomnia, the Creutzfeldt-Jakob disease (CJD), and Gerstmann-Sträussler-Scheinker syndrome (GSS) appear to be prion diseases. They result in progressive degeneration of the brain and eventual death. Mad cow disease has reached epidemic proportions in Great Britain and initially spread because cattle were fed bone meal made from cattle. It has now been shown that eating meat from cattle with BSE can cause a variant of Creutzfeldt-Jacob disease in humans. More than 90 people already have died in the United Kingdom and France from this source. CJD and GSS are rare and cosmopolitan in distribution among middle-aged people, while kuru has been found only in the Fore, an eastern New Guinea tribe. This tribe had a custom of consuming dead kinsmen, and women were given the honor of preparing the brain and practicing this ritual cannibalism. They and their children were infected by handling the diseased brain tissue. (Now that cannibalism has been eliminated, the incidence of kuru has decreased and is found only among the older adults.)

1. What are viroids and why are they of great interest?
2. How does a viroid differ from a virus?
3. What is a prion? In what way does a prion appear to differ fundamentally from viruses and viroids?

## Summary


1. Animal viruses are classified according to many properties; the most important are their morphology and nucleic acids (**figures 18.1–18.3**).
2. The first step in the animal virus life cycle is adsorption of the virus to a target cell receptor site; often special capsid structures are involved in this process.
3. Virus penetration of the host cell plasma membrane may be accompanied by capsid removal from the nucleic acid or uncoating. Most often penetration occurs through engulfment to form coated vesicles, but mechanisms such as direct fusion of the envelope with the plasma membrane also are employed (**figure 18.4**).
4. Early viral mRNA and proteins are involved in taking over the host cell and the synthesis of viral DNA and RNA. DNA replication usually takes place in the host nucleus and mRNA is initially manufactured by host enzymes.
5. Poxviruses differ from other DNA animal viruses in that DNA replication takes place in host cytoplasm and they carry an RNA polymerase. The parvoviruses are so small that they must conserve genome space by using overlapping genes and other similar mechanisms. Hepadnaviruses use reverse transcriptase to replicate their dsDNAgenome.
6. The genome of positive ssRNA viruses can act as an mRNA, whereas negative ssRNA virus genomes direct the synthesis of mRNA by a virus-associated transcriptase. Double-stranded RNA reoviruses use both virus-associated and newly synthesized transcriptases to make mRNA (**figure 18.6**).
7. RNA virus genomes are replicated in the host cell cytoplasm. Most ssRNA viruses use a viral replicase to synthesize a dsRNA replicative form that then directs the formation of new genomes.
8. Retroviruses use reverse transcriptase to synthesize a DNA copy of their RNA genome. After the double-stranded proviral DNA has been synthesized, it is integrated into the host genome and directs the formation of virus RNA and protein.
9. Late genes code for proteins needed in (a) capsid construction by a self-assembly process and (b) virus release.
10. Usually, naked virions are released upon cell lysis. In enveloped virus reproduction, virus release and envelope formation normally occur simultaneously after modification of the host plasma membrane, and the cell is not lysed.
11. Viruses can destroy host cells in many ways during cytocidal infections. These include such mechanisms as inhibition of host DNA, RNA, and protein synthesis; lysosomal damage; alteration of host cell membranes; and the formation of inclusion bodies.
12. Although most virus infections are acute or have a rapid onset and last for a relatively short time, some viruses can establish persistent infections lasting for years. Some infections are chronic. Viruses also can become dormant for a while and then resume activity in what is called a latent infection. Slow viruses may act so slowly that a disease develops over years.
13. Cancer is characterized by the formation of a malignant tumor that metastasizes or invades other tissues and can spread through the body. Carcinogenesis is a complex, multistep process involving many factors.
14. Viruses cause cancer in several ways. For example, they may bring a cancer-causing gene, or oncogene, into a cell, or the virion may insert a promoter next to a cellular oncogene and stimulate the gene to greater activity.

15. Most plant viruses have an RNA genome and may be either helical or icosahedral. Depending on the virus the RNA genome may be replicated by either a host RNA-dependent RNA polymerase or a virus-specific RNA replicase.
16. The TMV nucleocapsid forms spontaneously by self-assembly when disks of coat protein protomers complex with the RNA.
17. Plant viruses are transmitted in a variety of ways. Some enter through lesions in plant tissues, while others are transmitted by contaminated seeds, tubers, or pollen. Most are probably carried and inoculated by plant-feeding insects.
18. Mycoviruses from higher fungi have isometric capsids and dsRNA, whereas the viruses of lower fungi may have either dsRNA or dsDNA genomes.
19. Members of several virus families—most importantly the *Baculoviridae, Reoviridae,* and *Iridoviridae*—infect insects, and many of these viruses produce inclusion bodies that aid in their transmission.
20. Baculoviruses and other viruses are finding uses as biological control agents for insect pests.
21. Infectious agents simpler than viruses exist. For example, several plant diseases are caused by short strands of infectious RNA called viroids.
22. Prions are small agents associated with at least six degenerative nervous system disorders: scrapie, bovine spongiform encephalopathy, kuru, fatal familial insomnia, the Gerstmann-Sträussler-Scheinker syndrome, and Creutzfeldt-Jakob disease. The precise nature of prions is not yet clear.

## Key Terms

acute infection *410*
anaplasia *411*
cancer *411*
chronic virus infection *410*
coated vesicles *403*
cytocidal infection *410*
defective interfering (DI) particle *410*
early genes *403*
inclusion bodies *410*
late genes *408*
latent virus infection *410*
metastasis *411*
neoplasia *411*
oncogene *411*
persistent infection *410*
prion *416*
procapsids *408*
proviral DNA *407*
replicase *406*
replicative form *406*
retrovirus *407*
reverse transcriptase (RT) *407*
ribonuclease H *407*
slow virus diseases *410*
transcriptase *406*
tumor *411*
viroid *416*

## Questions for Thought and Review

1. Make a list of all the ways you can think of in which viruses differ from their eucaryotic host cells.
2. Compare animal and plant viruses in terms of entry into host cells. Why do they differ so greatly in this regard?
3. Compare lysogeny with the reproductive strategy of retroviruses. What advantage might there be in having one's genome incorporated into that of the host cell?
4. Inclusion bodies are mentioned more than once in the chapter. Where are they found, and what are their functions, if any?
5. Would it be advantageous for a virus to damage host cells? If not, why isn't damage to the host avoided? Is it possible that a virus might become less pathogenic when it has been associated with the host population for a longer time?
6. From what you know about cancer, is it likely that a single type of treatment can be used to cure it? What approaches might be effective in preventing cancer?
7. What is unusual about prions compared with viruses or bacterial pathogens that makes them so difficult to detect and treat?

## Critical Thinking Questions

1. Consider each of the following viral reproduction steps: adsorption, penetration, replication, and transcription. Suggest a strategy by which one could pharmacologically inhibit or discourage entry and propagation of viruses in animal cells. Can you explain host range using some of the same rationale?
2. How does one prove that a virus is causing cancer? Try to think of approaches other than those discussed in the chapter. Give a major reason why it is so difficult to prove that a specific virus causes human cancer. Is it accurate to say that viruses cause cancer? Explain why or why not.
3. Propose some experiments that might be useful in determining what prions are and how they cause disease.

## Additional Reading

Chapter 16 references also should be consulted, particularly the introductory and advanced texts.

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# PART VII

## The Diversity of the Microbial World



# CHAPTER 19

# Microbial Taxonomy

The stromatolites shown here are layered rocks formed by incorporation of minerals into microbial mats. Fossilized stromatolites indicate that microorganisms existed early in Earth's history.

## Outline

# Concepts

1. In order to make sense of the diversity of organisms, it is necessary to group similar organisms together and organize these groups in a nonoverlapping hierarchical arrangement. Taxonomy is the science of biological classification.
2. It appears that the procaryotic groups (Archaea and Bacteria) first developed, then the eucaryotes. This resulted in three domains: *Bacteria, Archaea,* and *Eucarya.* These domains differ from one another in rRNA sequences and many other ways.
3. The basic taxonomic group is the species, which is defined in terms of either sexual reproduction or general similarity.
4. Classifications are based on an analysis of possible evolutionary relationships (phylogenetic or phyletic classification) or on overall similarity (phenetic classification). The results of these analyses are often summarized in treelike diagrams called dendrograms.
5. Morphological, physiological, metabolic, ecological, genetic, and molecular characteristics are all useful in taxonomy because they reflect the organization and activity of the genome. Nucleic acid sequences are probably the best indicators of microbial phylogeny and relatedness because nucleic acids are either the genetic material itself or the products of gene transcription.
6. The first edition of *Bergey's Manual of Systematic Bacteriology* was largely phenotypic and divided procaryotes into groups based on easily determined characteristics such as shape, Gram-staining properties, oxygen relationships, and motility. The second edition, with its five volumes, is phylogenetically organized and distributes procaryotes among two domains and 25 phyla.
7. Bacterial taxonomy is rapidly changing due to the acquisition of new data, particularly the use of molecular techniques such as the comparison of ribosomal RNA structure and chromosome sequences. This is leading to new phylogenetic classifications.

*What's in a name? that which we call a rose*
*By any other name would smell as sweet. . . .*

*W. Shakespeare*

One of the most fascinating and attractive aspects of the microbial world is its extraordinary diversity. It seems that almost every possible experiment in shape, size, physiology, and life-style has been tried. The seventh part of the text focuses on this diversity. Chapter 19 introduces the general principles of microbial taxonomy. This is followed by a five-chapter (20–24) survey of the most important procaryotic groups. Part VII ends with an extensive introduction to the major types of eucaryotic microorganisms: fungi, algae, and protozoa.

## 19.1 General Introduction and Overview

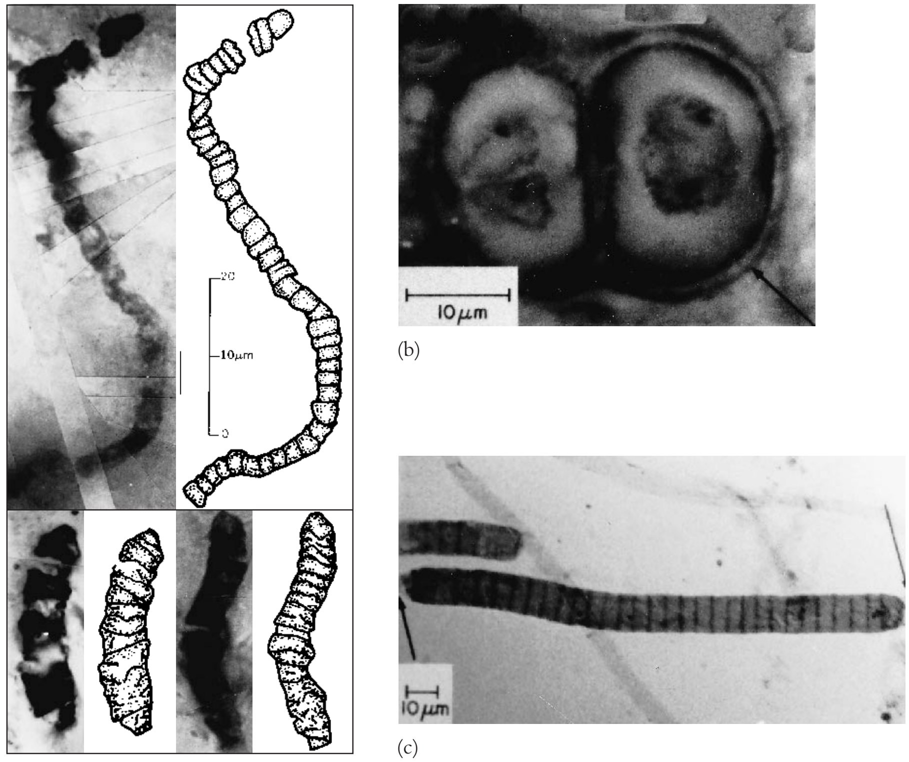
Because of the bewildering diversity of living organisms, it is desirable to classify or arrange them into groups based on their mutual similarities. **Taxonomy** [Greek *taxis,* arrangement or order, and *nomos,* law, or *nemein,* to distribute or govern] is defined as the science of biological classification. In a broader sense it consists of three separate but interrelated parts: classification, nomenclature, and identification. **Classification** is the arrangement of organisms into groups or **taxa** (s., **taxon**) based on mutual similarity or evolutionary relatedness. **Nomenclature** is the branch of taxonomy concerned with the assignment of names to taxonomic groups in agreement with published rules. **Identification** is the practical side of taxonomy, the process of determining that a particular isolate belongs to a recognized taxon.

People often think of taxonomy as trivial and boring, simply a matter of splitting hairs over names of organisms. Actually, taxonomy is important for several reasons. First, it allows us to organize huge amounts of knowledge about organisms because all members of a particular group share many characteristics. In a sense it is something like a giant filing system or library catalogue that provides easy access to information. The more accurate the classification, the more information-rich and useful it is. Second, taxonomy allows us to make predictions and frame hypotheses for further research based on knowledge of similar organisms. If a relative has some property, the microorganism in question also may have the same characteristic. Third, taxonomy places microorganisms in meaningful, useful groups with precise names so that microbiologists can work with them and communicate efficiently. Just as effective written communication is not possible without adequate vocabulary, correct spelling, and good grammar, microbiology is not possible without taxonomy. Fourth, taxonomy is essential for accurate identification of microorganisms. Its practical importance in this respect can hardly be overemphasized. For example, it is essential to clinical microbiology (*see chapter 36*); treatment often is exceptionally difficult when the pathogen is unknown.

The term **systematics** often is used for taxonomy. However, many taxonomists define it in more general terms as "the scientific study of organisms with the ultimate object of characterizing and arranging them in an orderly manner." Any study of the nature of organisms, when the knowledge gained is used in taxonomy, is a part of systematics. Thus it encompasses disciplines such as morphology, ecology, epidemiology, biochemistry, molecular biology, and physiology.

Microbial taxonomy is too broad a subject for adequate coverage in a single chapter. Therefore this chapter emphasizes general principles and uses examples primarily from procaryotic taxonomy. The taxonomy of each major eucaryotic microbial group is reviewed where the group is introduced in subsequent chapters.

Currently microbial taxonomy is in ferment because of the use of new molecular techniques in classifying microorganisms. Even though these new advances have generated much excitement and are drastically changing microbial taxonomy, more traditional approaches still have value and also will be outlined. The chapter begins with a general overview of taxonomy and then briefly describes the major characteristics used in microbial taxonomy. Next, the assessment of microbial phylogeny is discussed, followed by a description of the major divisions of life and *Bergey's Manual of Systematic Bacteriology.* The chapter ends with a brief survey of procaryotic phylogeny and diversity.



(a) (b) (c)

**Figure 19.1 Fossilized Bacteria.** Several microfossils resembling cyanobacteria are shown, some with interpretive drawings. (**a**) Thin sections of Archean Apex chert from Western Australia; the fossils are about 3.5 billion years old. (**b**) *Gloeodiniopsis,* about 1.5 billion years old, from carbonaceous chert in the Satka Formation of the southern Ural Mountains. The arrow points to the enclosing sheath. (**c**) *Palaeolyngbya,* about 950 million years old, from carbonaceous shale of the Lakhanda Formation of the Khabarovsk region in eastern Siberia.

## 19.2 Microbial Evolution and Diversity

It has been estimated that our planet is about 4.6 billion years old. Fossilized remains of procaryotic cells around 3.5 to 3.8 billion years old have been discovered in stromatolites and sedimentary rocks (**figure 19.1**). **Stromatolites** are layered or stratified rocks, often domed, that are formed by incorporation of mineral sediments into microbial mats (**figure 19.2**). Modern stromatolites are formed by cyanobacteria; presumably at least some fossilized stromatolites were formed in the same way. Thus procaryotic life arose very shortly after the earth cooled. Very likely the earliest procaryotes were anaerobic. Cyanobacteria and oxygen-producing photosynthesis probably developed 2.5 to 3.0 billion or more years ago. Microbial diversity increased greatly as oxygen became more plentiful.

The studies of Carl Woese and his collaborators on rRNA sequences in procaryotic cells suggest that procaryotes divided into two distinct groups very early on. **Figure 19.3** depicts a universal phylogenetic tree that reflects these views. The tree is divided into three major branches representing the three primary

**Figure 19.2 Stromatolites.** These are stromatolites at Shark Bay, Western Australia. Modern stromatolites are layered or stratified rocks formed by the incorporation of calcium sulfates, calcium carbonates, and other minerals into microbial mats. The mats are formed by cyanobacteria and other microorganisms.

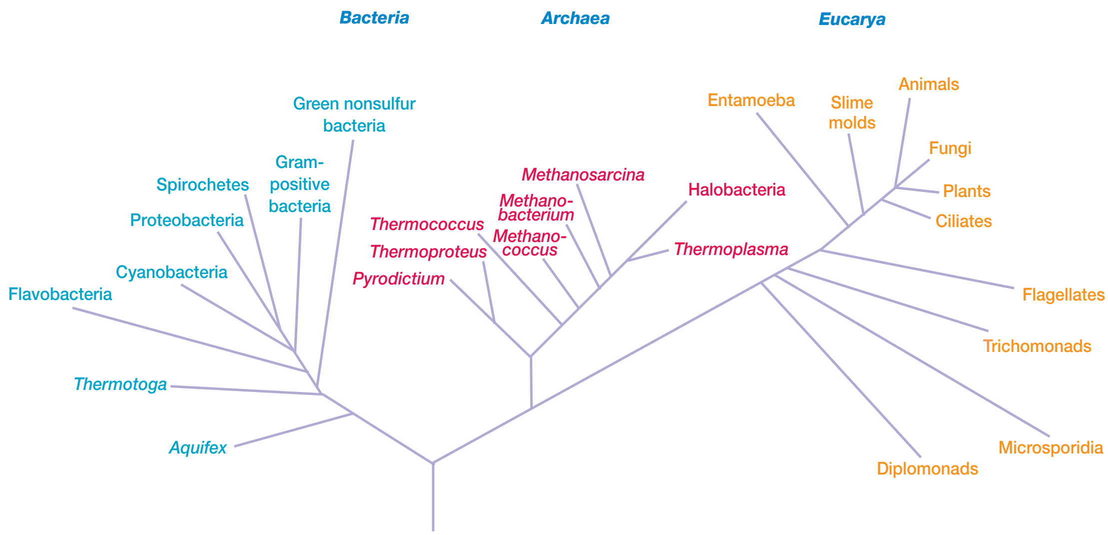



**Figure 19.3** **Universal Phylogenetic Tree.** These relationships were determined from rRNA sequence comparisons. *Source: Adapted from G. J. Olsen and C. R. Woese. "Ribosomal RNA: A Key to Phylogeny" in* The FASEB Journal, *7:113–123, 1993.*

groups: *Bacteria, Archaea,* and *Eucarya.* The archaea and bacteria first diverged, then the eucaryotes developed. These three primary groups are called **domains** and placed above the phylum and kingdom levels (the traditional kingdoms are distributed among these three domains). The domains differ markedly from one another. Eucaryotic organisms with primarily glycerol fatty acyl diester membrane lipids and eucaryotic rRNA belong to the ***Eucarya.*** The domain ***Bacteria*** contains procaryotic cells with bacterial rRNA and membrane lipids that are primarily diacyl glycerol diesters. Procaryotes having isoprenoid glycerol diether or diglycerol tetraether lipids (*see pp. 452–53*) in their membranes and archaeal rRNA compose the third domain, ***Archaea.***

It appears likely that modern eucaryotic cells arose from procaryotes about 1.4 billion years ago. There has been considerable speculation about how eucaryotes might have developed from procaryotic ancestors. It is not certain how this process occurred, and two hypotheses have been proposed. According to the first, nuclei, mitochondria, and chloroplasts arose by invagination of the plasma membrane to form double-membrane structures containing genetic material and capable of further development and specialization. The similarities between chloroplasts, mitochondria, and modern bacteria are due to conservation of primitive procaryotic features by the slowly changing organelles.

According to the more popular **endosymbiotic hypothesis,** the first event was nucleus formation in the pro-eucaryotic cell. The ancestral eucaryotic cell may have developed from a fusion of ancient bacteria and archaea. Possibly a gram-negative bacterial host cell that had lost its cell wall engulfed an archaeon to form an endosymbiotic association. The archaeon subsequently lost its wall and plasma membrane, while the host bacterium developed membrane infolds. Eventually the host genome was transferred to the original archaeon, and a nucleus and the endoplasmic reticulum was formed. Both bacterial and archaeal genes could be lost during formation of the eucaryotic genome. It should be noted that many believe that the *Archaea* and *Eucarya* are more closely related than this hypothetical scenario implies. They propose that the eucaryotic line diverged from the *Archaea* and then the nucleus formed, possibly from the Golgi apparatus.

Mitochondria and chloroplasts appear to have developed later. The free-living, fermenting ancestral eucaryote with its nucleus established a permanent symbiotic relationship with photosynthetic bacteria, which then evolved into chloroplasts. Cyanobacteria have been considered the most likely ancestors of chloroplasts. More recently *Prochloron* has become the favorite candidate. *Prochloron* (*see pp. 475–76*) lives within marine invertebrates and resembles chloroplasts in containing both chlorophyll *a* and *b,* but not phycobilins. The existence of this bacterium suggests that chloroplasts arose from a common ancestor of prochlorophytes and cyanobacteria. Mitochondria arose from an endosymbiotic relationship between the free-living primitive eucaryote and bacteria with aerobic respiration (possibly an ancestor of three modern groups: *Agrobacterium, Rhizobium,* and *Rickettsia*). Some have proposed that aerobic respiration actually arose before oxygenic (oxygen-producing) photosynthesis and made use of small amounts of oxygen available at this early stage of planetary development. The exact sequence of development is still unclear.

The endosymbiotic hypothesis has received support from the discovery of an endosymbiotic cyanobacterium that inhabits the biflagellate protist *Cyanophora paradoxa* and acts as its chloroplast. This endosymbiont, called a cyanelle, resembles the cyanobacteria in its photosynthetic pigment system and fine structure and it is surrounded by a peptidoglycan layer. It differs

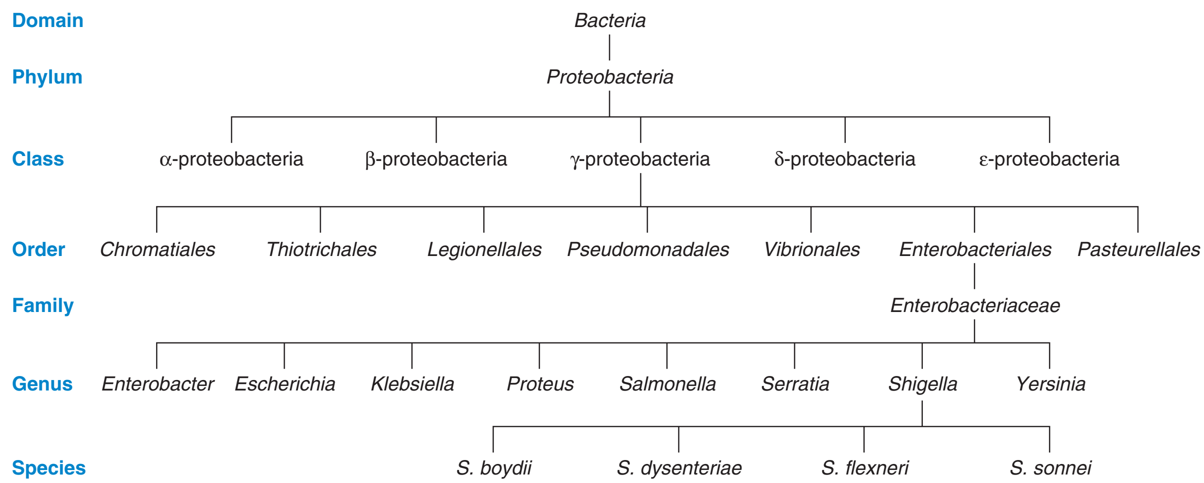



**Figure 19.4 Hierarchical Arrangement in Taxonomy.** In this example, members of the genus *Shigella* are placed within higher taxonomic ranks. Not all classification possibilies are given for each rank to simplify the diagram.

from cyanobacteria in lacking the lipopolysaccharide outer membrane characteristic of gram-negative bacteria. The cyanelle may be a recently established endosymbiont that is evolving into a chloroplast. Further support is provided by rRNA trees, which locate chloroplast RNA within the cyanobacteria.

At present both hypotheses have supporters. It is possible that new data may help resolve the issue to everyone's satisfaction. However, these hypotheses concern processes that occurred in the distant past and cannot be directly observed. Thus a complete consensus on the matter may never be reached.

**Table 19.1 An Example of Taxonomic Ranks and Names**

| Rank | Example |
|---|---|
| Domain | *Bacteria* |
| Phylum | *Proteobacteria* |
| Class | γ-Proteobacteria |
| Order | *Enterobacteriales* |
| Family | *Enterobacteriaceae* |
| Genus | *Shigella* |
| Species | *S. dysenteriae* |

## 19.3 Taxonomic Ranks

In preparing a classification scheme, one places the microorganism within a small, homogeneous group that is itself a member of larger groups in a nonoverlapping hierarchical arrangement. A category in any rank unites groups in the level below it based on shared properties (**figure 19.4**). In procaryotic taxonomy the most commonly used levels or ranks (in ascending order) are species, genera, families, orders, classes, and phyla. Microbial groups at each level or rank have names with endings or suffixes characteristic of that level (**table 19.1**). Microbiologists often use informal names in place of formal hierarchical ones. Typical examples of such names are purple bacteria, spirochetes, methane-oxidizing bacteria, sulfate-reducing bacteria, and lactic acid bacteria.

The basic taxonomic group in microbial taxonomy is the **species.** Taxonomists working with higher organisms define the term species differently than do microbiologists. Species of higher organisms are groups of interbreeding or potentially interbreeding natural populations that are reproductively isolated from other groups. This is a satisfactory definition for organisms capable of sexual reproduction but fails with many microorganisms because they do not reproduce sexually. Procaryotic species are characterized by phenotypic and genotypic (*see chapter 11*) differences. A **procaryotic species** is a collection of strains that share many stable properties and differ significantly from other groups of strains. This definition is very subjective and can be interpreted in many ways. The following more precise definition has been proposed by some bacterial taxonomists. A species (genomospecies) is a collection of strains that have a similar G + C composition and 70% or greater similarity as judged by DNA hybridization experiments (pp. 429–32). Ideally a species also should be phenotypically distinguishable from other similar species. A **strain** is a population of organisms that is distinguishable from at least some other populations within a particular taxonomic category. It is considered to have descended from a single organism or pure culture isolate. Strains within a species may differ slightly from one another in many ways. **Biovars** are variant procaryotic strains characterized by biochemical or physiological differences, **morphovars** differ morphologically, and **serovars** have distinctive antigenic properties. One strain of a species is designated as the **type strain.** It is usually one of the first strains studied and often is more fully characterized than other strains; however, it does not have to be the most representative member. The type strain for the species is called the type species and is the nomenclatural type or the holder of the species name. A nomenclatural type is a device to ensure fixity of names when taxonomic rearrangements take place. For example, the type species must remain within the genus of which it is the nomenclatural type. Only those strains very similar to the type strain or type species are included in a species. Each species is assigned to a genus, the next rank in the taxonomic hierarchy. A **genus** is a well-defined group of one or more species that is clearly separate from other genera. In practice there is considerable subjectivity in assigning species to a genus, and taxonomists may disagree about the composition of genera.

Microbiologists name microorganisms by using the **binomial system** of the Swedish botanist Carl von Linné, or Carolus Linnaeus as he often is called. The Latinized, italicized name consists of two parts. The first part, which is capitalized, is the generic name, and the second is the uncapitalized specific epithet (e.g., *Escherichia coli*). The specific epithet is stable; the oldest epithet for a particular organism takes precedence and must be used. In contrast, a generic name can change if the organism is assigned to another genus because of new information. For example, the genus *Streptococcus* has been divided to form two new genera, *Enterococcus* and *Lactococcus* based on rRNA analysis and other characteristics. Thus *Streptococcus faecalis* is now *Enterococcus faecalis.* Often the name will be shortened by abbreviating the genus name with a single capital letter, for example *E. coli.* Approved lists of bacterial names were published in 1980 in the *International Journal of Systematic Bacteriology,* and new valid names are published periodically. *Bergey's Manual of Systematic Bacteriology* contains the currently accepted system of procaryotic taxonomy and will be discussed later in the chapter.

1. Define the following terms: taxonomy, classification, taxon, nomenclature, identification, systematics, species, strain, type strain, and binomial system.
2. Briefly describe the three domains into which living organisms may be divided.
3. How might the eucaryotic cell have arisen according to the endosymbiotic hypothesis?
4. How does the definition of species differ between organisms that reproduce sexually and those not able to do so?

## 19.4 Classification Systems

Once taxonomically relevant characteristics of microorganisms have been collected, they may be used to construct a classification system. The most desirable classification system, called a **natural classification,** arranges organisms into groups whose members share many characteristics and reflects as much as possible the biological nature of organisms. Linnaeus developed the first natural classification, based largely on anatomical characteristics, in the middle of the eighteenth century. It was a great improvement over previously employed artificial systems because knowledge of an organism's position in the scheme provided information about many of its properties. For example, classification of humans as mammals denotes that they have hair, self-regulating body temperature, and milk-producing mammary glands in the female.

There are two general ways in which classification systems can be constructed. Organisms can be grouped together based on overall similarity to form a phenetic system or they can be grouped based on probable evolutionary relationships to produce a phylogenetic system. Computers may be used to analyze data for the production of phenetic classifications. The process is called numerical taxonomy. This section briefly discusses phenetic and phylogenetic classifications, and describes numerical taxonomy.

### Phenetic Classification

Many taxonomists maintain that the most natural classification is the one with the greatest information content or predictive value. A good classification should bring order to biological diversity and may even clarify the function of a morphological structure. For example, if motility and flagella are always associated in particular microorganisms, it is reasonable to suppose that flagella are involved in at least some types of motility. When viewed in this way, the best natural classification system may be a **phenetic system,** one that groups organisms together based on the mutual similarity of their phenotypic characteristics. Although phenetic studies can reveal possible evolutionary relationships, they are not dependent on phylogenetic analysis. They compare many traits without assuming that any features are more phylogenetically important than others—that is, unweighted traits are employed in estimating general similarity. Obviously the best phenetic classification is one constructed by comparing as many attributes as possible. Organisms sharing many characteristics make up a single group or taxon.

### Numerical Taxonomy

The development of computers has made possible the quantitative approach known as **numerical taxonomy.** Peter H. A. Sneath and Robert Sokal have defined numerical taxonomy as "the grouping by numerical methods of taxonomic units into taxa on the basis of their character states." Information about the properties of organisms is converted into a form suitable for numerical analysis and then compared by means of a computer. The resulting classification is based on general similarity as judged by comparison of many characteristics, each given equal weight. This approach was not feasible before the advent of computers because of the large number of calculations involved.

The process begins with a determination of the presence or absence of selected characters in the group of organisms under study. A character usually is defined as an attribute about which a single statement can be made. Many characters, at least 50 and preferably several hundred, should be compared for an accurate and reliable classification. It is best to include many different kinds of data: morphological, biochemical, and physiological.

After character analysis, an association coefficient, a function that measures the agreement between characters possessed by two organisms, is calculated for each pair of organisms in the group. The **simple matching coefficient** ($S_{SM}$), the most commonly used coefficient in bacteriology, is the proportion of

## Table 19.2 The Calculation of Association Coefficients for Two Organisms

In this example, organisms A and B are compared in terms of the characters they do and do not share. The terms in the association coefficient equations are defined as follows:

| | | Organism B | |
|---|---|---|---|
| | | 1 | 0 |
| Organism A | 1 | *a* | *b* |
| | 0 | *c* | *d* |

*a* = number of characters coded as present (1) for both organisms
*b* and *c* = numbers of characters differing (1,0 or 0,1) between the two organisms
*d* = number of characters absent (0) in both organisms
Total number of characters compared = $a + b + c + d$

The simple matching coefficient ($S_{SM}$) = $\dfrac{a + d}{a + b + c + d}$

The Jaccard coefficient ($S_J$) = $\dfrac{a}{a + b + c}$

characters that match regardless of whether the attribute is present or absent (**table 19.2**). Sometimes the **Jaccard coefficient** ($S_J$) is calculated by ignoring any characters that both organisms lack (table 19.2). Both coefficients increase linearly in value from 0.0 (no matches) to 1.0 (100% matches).

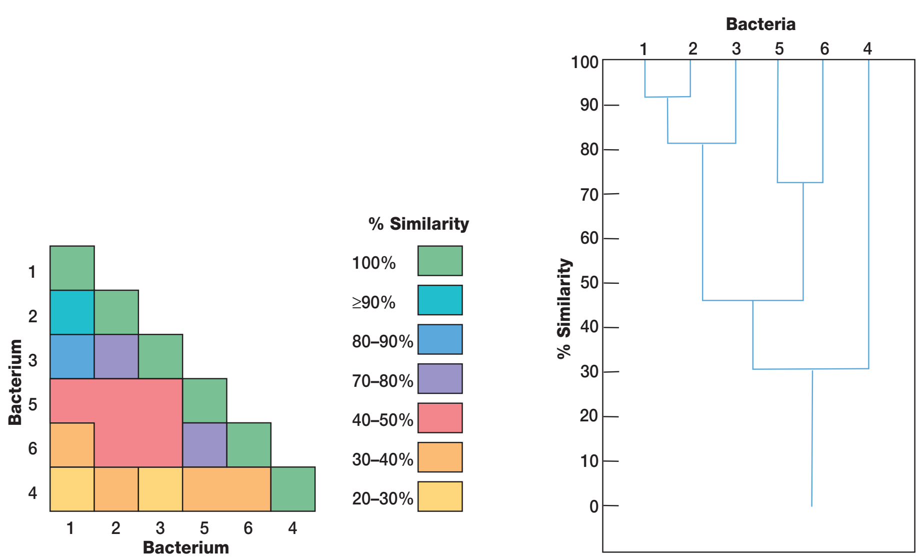
The simple matching coefficients, or other association coefficients, are then arranged to form a **similarity matrix.** This is a matrix in which the rows and columns represent organisms, and each value is an association coefficient measuring the similarity of two different organisms so that each organism is compared to every other one in the table (**figure 19.5*a***). Organisms with great similarity are grouped together and separated from dissimilar organisms (figure 19.5*b*); such groups of organisms are called **phenons** (sometimes called phenoms).

The results of numerical taxonomic analysis are often summarized with a treelike diagram called a **dendrogram** (figure 19.5*c*). The diagram usually is placed on its side with the X-axis or abscissa graduated in units of similarity. Each branch point is at the similarity value relating the two branches. The organisms in the two branches share so many characteristics that the two groups are seen to be separate only after examination of association coefficients greater than the magnitude of the branch point value. Below the branch point value, the two groups appear to be one. The ordinate in such a dendrogram has no special significance, and the clusters may be arranged in any convenient order.

| Bacterium | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 1.0 | | | | | |
| 2 | 0.92 | 1.0 | | | | |
| 3 | 0.81 | 0.77 | 1.0 | | | |
| 4 | 0.27 | 0.31 | 0.29 | 1.0 | | |
| 5 | 0.43 | 0.41 | 0.45 | 0.30 | 1.0 | |
| 6 | 0.38 | 0.42 | 0.44 | 0.32 | 0.72 | 1.0 |
| | 1 | 2 | 3 | 4 | 5 | 6 |

Bacterium

(a)




(b) (c)

**Figure 19.5 Clustering and Dendrograms in Numerical Taxonomy.** (**a**) A small similarity matrix that compares six strains of bacteria. The degree of similarity ranges from none (0.0) to complete similarity (1.0). (**b**) The bacteria have been rearranged and joined to form clusters of similar strains. For example, strains 1 and 2 are the most similar. The cluster of 1 plus 2 is fairly similar to strain 3, but not at all to strain 4. (**c**) A dendrogram showing the results of the analysis in part *b.* Strains 1 and 2 are members of a 90-phenon, and strains 1–3 form an 80-phenon. While strains 1–3 may be members of a single species, it is quite unlikely that strains 4–6 belong to the same species as 1–3.

The significance of these clusters or phenons in traditional taxonomic terms is not always evident, and the similarity levels at which clusters are labeled species, genera, and so on, are a matter of judgment. Sometimes groups are simply called phenons and preceded by a number showing the similarity level above which they appear (e.g., a 70-phenon is a phenon with 70% or greater similarity among its constituents). Phenons formed at about 80% similarity often are equivalent to species.

Numerical taxonomy has already proved to be a powerful tool in microbial taxonomy. Although it often has simply reconfirmed already existing classification schemes, sometimes accepted classifications are found wanting. Numerical taxonomic methods also can be used to compare sequences of macromolecules such as RNA and proteins.

### Phylogenetic Classification

Following the publication in 1859 of Darwin's *On the Origin of Species,* biologists began trying to develop **phylogenetic** or **phyletic classification systems.** These are systems based on evolutionary relationships rather than general resemblance (the term **phylogeny** [Greek *phylon,* tribe or race, and *genesis,* generation or origin] refers to the evolutionary development of a species). This has proven difficult for procaryotes and other microorganisms, primarily because of the lack of a good fossil record. The direct comparison of genetic material and gene products such as RNA and proteins overcomes many of these problems.

1. What is a natural classification?
2. What are phylogenetic (phyletic) and phenetic classification systems? How do the two systems differ?
3. What is numerical taxonomy and why are computers so important to this approach?
4. Define the following terms: association coefficient, simple matching coefficient, Jaccard coefficient, similarity matrix, phenon, and dendrogram.
5. Which pair of species has more mutual similarity, a pair with an association coefficient or 0.9 or one with a coefficient of 0.6? Why?

## 19.5 Major Characteristics Used in Taxonomy

Many characteristics are used in classifying and identifying microorganisms. This section briefly reviews some of the most taxonomically important properties. For sake of clarity, characteristics have been divided into two groups: classical and molecular. Methods often employed in routine laboratory identification of bacteria are covered in the chapter on clinical microbiology (*see chapter 36*).

### Classical Characteristics

Classical approaches to taxonomy make use of morphological, physiological, biochemical, ecological, and genetic characteristics. These characteristics have been employed in microbial taxonomy for many years. They are quite useful in routine identification and may provide phylogenetic information as well.

**Table 19.3 Some Morphological Features Used in Classification and Identification**

| Feature | Microbial Groups |
|---|---|
| Cell shape | All major groups[a] |
| Cell size | All major groups |
| Colonial morphology | All major groups |
| Ultrastructural characteristics | All major groups |
| Staining behavior | Bacteria, some fungi |
| Cilia and flagella | All major groups |
| Mechanism of motility | Gliding bacteria, spirochetes |
| Endospore shape and location | Endospore-forming bacteria |
| Spore morphology and location | Bacteria, algae, fungi |
| Cellular inclusions | All major groups |
| Color | All major groups |

[a]Used in classifying and identifying at least some bacteria, algae, fungi, and protozoa.

#### Morphological Characteristics

Morphological features are important in microbial taxonomy for many reasons. Morphology is easy to study and analyze, particularly in eucaryotic microorganisms and the more complex procaryotes. In addition, morphological comparisons are valuable because structural features depend on the expression of many genes, are usually genetically stable, and normally (at least in eucaryotes) these do not vary greatly with environmental changes. Thus morphological similarity often is a good indication of phylogenetic relatedness.

Many different morphological features are employed in the classification and identification of microorganisms (**table 19.3**). Although the light microscope has always been a very important tool, its resolution limit of about 0.2 μm (*see chapter 2*) reduces its usefulness in viewing smaller microorganisms and structures. The transmission and scanning electron microscopes, with their greater resolution, have immensely aided the study of all microbial groups.

#### Physiological and Metabolic Characteristics

Physiological and metabolic characteristics are very useful because they are directly related to the nature and activity of microbial enzymes and transport proteins. Since proteins are gene products, analysis of these characteristics provides an indirect comparison of microbial genomes. **Table 19.4** lists some of the most important of these properties.

#### Ecological Characteristics

Many properties are ecological in nature since they affect the relation of microorganisms to their environment. Often these are taxonomically valuable because even very closely related microorganisms can differ considerably with respect to ecological characteristics. Microorganisms living in various parts of the human body markedly

**Table 19.4 Some Physiological and Metabolic Characteristics Used in Classification and Identification**

| |
|---|
| Carbon and nitrogen sources |
| Cell wall constituents |
| Energy sources |
| Fermentation products |
| General nutritional type |
| Growth temperature optimum and range |
| Luminescence |
| Mechanisms of energy conversion |
| Motility |
| Osmotic tolerance |
| Oxygen relationships |
| pH optimum and growth range |
| Photosynthetic pigments |
| Salt requirements and tolerance |
| Secondary metabolites formed |
| Sensitivity to metabolic inhibitors and antibiotics |
| Storage inclusions |

differ from one another and from those growing in freshwater, terrestrial, and marine environments. Some examples of taxonomically important ecological properties are life cycle patterns; the nature of symbiotic relationships; the ability to cause disease in a particular host; and habitat preferences such as requirements for temperature, pH, oxygen, and osmotic concentration. Many growth requirements are also considered physiological characteristics (table 19.4).

### Genetic Analysis

Because most eucaryotes are able to reproduce sexually, genetic analysis has been of considerable usefulness in the classification of these organisms. As mentioned earlier, the species is defined in terms of sexual reproduction where possible. Although procaryotes do not reproduce sexually, the study of chromosomal gene exchange through transformation and conjugation is sometimes useful in their classification.

Transformation can occur between different procaryotic species but only rarely between genera. The demonstration of transformation between two strains provides evidence of a close relationship since transformation cannot occur unless the genomes are fairly similar. Transformation studies have been carried out with several genera: *Bacillus, Micrococcus, Haemophilus, Rhizobium,* and others. Despite transformation's usefulness, its results are sometimes hard to interpret because an absence of transformation may result from factors other than major differences in DNA sequence. Transformation (pp. 228, 305–7); Conjugation (pp. 302–5)

Conjugation studies also yield taxonomically useful data, particularly with the enteric bacteria (*see section 22.3*). For example, *Escherichia* can undergo conjugation with the genera *Salmonella* and *Shigella* but not with *Proteus* and *Enterobacter.* These observations fit with other data showing that the first three of these genera are more closely related to one another than to *Proteus* and *Enterobacter.*

Plasmids (*see section 13.2*) are undoubtedly important in taxonomy because they are present in most bacterial genera, and many carry genes coding for phenotypic traits. Because plasmids could have a significant effect on classification if they carried the gene for a trait of major importance in the classification scheme, it is best to base a classification on many characters. When the identification of a group is based on a few characteristics and some of these are coded for by plasmid genes, errors may result. For example, hydrogen sulfide production and lactose fermentation are very important in the taxonomy of the enteric bacteria, yet genes for both traits can be borne on plasmids as well as bacterial chromosomes. One must take care to avoid errors as a result of plasmid-borne traits.

## Molecular Characteristics

Some of the most powerful approaches to taxonomy are through the study of proteins and nucleic acids. Because these are either direct gene products or the genes themselves, comparisons of proteins and nucleic acids yield considerable information about true relatedness. These more recent molecular approaches have become increasingly important in procaryotic taxonomy.

### Comparison of Proteins

The amino acid sequences of proteins are direct reflections of mRNA sequences and therefore closely related to the structures of the genes coding for their synthesis. For this reason, comparisons of proteins from different microorganisms are very useful taxonomically. There are several ways to compare proteins. The most direct approach is to determine the amino acid sequence of proteins with the same function. The sequences of proteins with dissimilar functions often change at different rates; some sequences change quite rapidly, whereas others are very stable. Nevertheless, if the sequences of proteins with the same function are similar, the organisms possessing them are probably closely related. The sequences of cytochromes and other electron transport proteins, histones, heat-shock proteins, transcription and translation proteins, and a variety of metabolic enzymes have been used in taxonomic studies. Because protein sequencing is slow and expensive, more indirect methods of comparing proteins frequently have been employed. The electrophoretic mobility of proteins (*see pp. 327–28*) is useful in studying relationships at the species and subspecies levels. Antibodies can discriminate between very similar proteins, and immunologic techniques are used to compare proteins from different microorganisms. Antibody-antigen reactions in vitro (pp. 774–84)

The physical, kinetic, and regulatory properties of enzymes have been employed in taxonomic studies. Because enzyme behavior reflects amino acid sequence, this approach is useful in studying some microbial groups, and group-specific patterns of regulation have been found.

### Nucleic Acid Base Composition

Microbial genomes can be directly compared, and taxonomic similarity can be estimated in many ways. The first, and possibly the simplest, technique to be employed is the determination of DNA base composition. DNA contains four purine and pyrimidine bases: adenine (A), guanine (G), cytosine (C), and thymine (T). In double-stranded DNA, A pairs with T, and G pairs with C. Thus the (G + C)/(A + T) ratio or **G + C content,** the percent of

G + C in DNA, reflects the base sequence and varies with sequence changes as follows: DNA and RNA structure (pp 230–35)

$$\text{Mol\% G + C} = \frac{G + C}{G + C + A + T} \times 100$$

The base composition of DNA can be determined in several ways. Although the G + C content can be ascertained after hydrolysis of DNA and analysis of its bases with high-performance liquid chromatography (HPLC), physical methods are easier and more often used. The G + C content often is determined from the **melting temperature** ($T_m$) of DNA. In double-stranded DNA three hydrogen bonds join GC base pairs, and two bonds connect AT base pairs (*see section 11.2*). As a result DNA with a greater G + C content will have more hydrogen bonds, and its strands will separate only at higher temperatures—that is, it will have a higher melting point. DNA melting can be easily followed spectrophotometrically because the absorbance of 260 nm UV light by DNA increases during strand separation. When a DNA sample is slowly heated, the absorbance increases as hydrogen bonds are broken and reaches a plateau when all the DNA has become single stranded (**figure 19.6**). The midpoint of the rising curve gives the melting temperature, a direct measure of the G + C content. Since the density of DNA also increases linearly with G + C content, the percent G + C can be obtained by centrifuging DNA in a CsCl density gradient (*see chapter 16*).

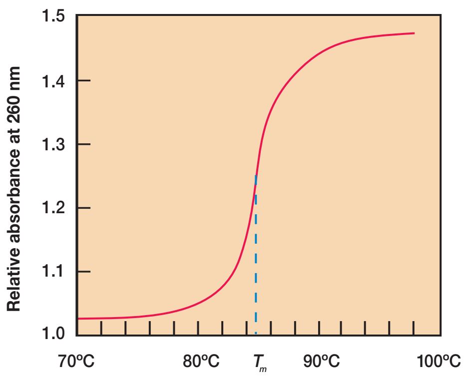



**Figure 19.6 A DNA Melting Curve.** The $T_m$ is indicated.

The G + C content of many microorganisms has been determined (**table 19.5**). The G + C content of DNA from animals and higher plants averages around 40% and ranges between 30 and 50%. In contrast, the DNA of both eucaryotic and procaryotic microorganisms varies greatly in G + C content; procaryotic G + C

## Table 19.5 Representative G + C Contents of Microorganisms

| Organism | Percent G + C |
|---|---|
| **Bacteria** | |
| *Actinomyces* | 59–73 |
| *Anabaena* | 38–44 |
| *Bacillus* | 32–62 |
| *Bacteroides* | 28–61 |
| *Bdellovibrio* | 33–52 |
| *Caulobacter* | 63–67 |
| *Chlamydia* | 41–44 |
| *Chlorobium* | 49–58 |
| *Chromatium* | 48–70 |
| *Clostridium* | 21–54 |
| *Cytophaga* | 33–42 |
| *Deinococcus* | 62–70 |
| *Escherichia* | 48–52 |
| *Halobacterium* | 66–68 |
| *Hyphomicrobium* | 59–67 |
| *Methanobacterium* | 32–50 |
| *Micrococcus* | 64–75 |
| *Mycobacterium* | 62–70 |
| *Mycoplasma* | 23–40 |
| *Myxococcus* | 68–71 |
| *Neisseria* | 47–54 |
| *Nitrobacter* | 60–62 |
| *Oscillatoria* | 40–50 |
| *Prochloron* | 41 |
| *Proteus* | 38–41 |
| *Pseudomonas* | 58–70 |
| *Rhodospirillum* | 62–66 |
| *Rickettsia* | 29–33 |
| *Salmonella* | 50–53 |
| *Spirillum* | 38 |
| *Spirochaeta* | 51–65 |
| *Staphylococcus* | 30–38 |
| *Streptococcus* | 33–44 |
| *Streptomyces* | 69–73 |
| *Sulfolobus* | 31–37 |
| *Thermoplasma* | 46 |
| *Thiobacillus* | 52–68 |
| *Treponema* | 25–54 |
| **Algae** | |
| *Acetabularia mediterranea* | 37–53 |
| *Chlamydomonas* | 60–68 |
| *Chlorella* | 43–79 |
| *Cyclotella cryptica* | 41 |
| *Euglena gracilis* | 46–55 |
| *Nitella* | 49 |
| *Nitzschia angularis* | 47 |
| *Ochromonas danica* | 48 |
| *Peridinium triquetrum* | 53 |
| *Scenedesmus* | 52–64 |
| *Spirogyra* | 39 |
| *Volvox carteri* | 50 |
| **Protozoa** | |
| *Acanthamoeba castellanii* | 56–58 |
| *Amoeba proteus* | 66 |
| *Paramecium* spp. | 29–39 |
| *Plasmodium berghei* | 41 |
| *Stentor polymorphus* | 45 |
| *Tetrahymena* | 19–33 |
| *Trichomonas* | 29–34 |
| *Trypanosoma* | 45–59 |
| **Slime Molds** | |
| *Dictyostelium* | 22–25 |
| *Lycogala* | 42 |
| *Physarum polycephalum* | 38–42 |
| **Fungi** | |
| *Agaricus bisporus* | 44 |
| *Amanita muscaria* | 57 |
| *Aspergillus niger* | 52 |
| *Blastocladiella emersonii* | 66 |
| *Candida albicans* | 33–35 |
| *Claviceps purpurea* | 53 |
| *Coprinus lagopus* | 52–53 |
| *Fomes fraxineus* | 56 |
| *Mucor rouxii* | 38 |
| *Neurospora crassa* | 52–54 |
| *Penicillium notatum* | 52 |
| *Polyporus palustris* | 56 |
| *Rhizopus nigricans* | 47 |
| *Saccharomyces cerevisiae* | 36–42 |
| *Saprolegnia parasitica* | 61 |

content is the most variable, ranging from around 25 to almost 80%. Despite such a wide range of variation, the G + C content of strains within a particular species is constant. If two organisms differ in their G + C content by more than about 10%, their genomes have quite different base sequences. On the other hand, it is not safe to assume that organisms with very similar G + C contents also have similar DNA base sequences because two very different base sequences can be constructed from the same proportions of AT and GC base pairs. Only if two microorganisms also are alike phenotypically does their similar G + C content suggest close relatedness.

G + C content data are taxonomically valuable in at least two ways. First, they can confirm a taxonomic scheme developed using other data. If organisms in the same taxon are too dissimilar in G + C content, the taxon probably should be divided. Second, G + C content appears to be useful in characterizing procaryotic genera since the variation within a genus is usually less than 10% even though the content may vary greatly between genera. For example, *Staphylococcus* has a G + C content of 30 to 38%, whereas *Micrococcus* DNA has 64 to 75% G + C; yet these two genera of gram-positive cocci have many other features in common.

### Nucleic Acid Hybridization

The similarity between genomes can be compared more directly by use of **nucleic acid hybridization** studies. If a mixture of single-stranded DNA formed by heating dsDNA is cooled and held at a temperature about 25°C below the $T_m$, strands with complementary base sequences will reassociate to form stable dsDNA, whereas noncomplementary strands will remain single (**figure 19.7**). Because strands with similar, but not identical, sequences associate to form less temperature stable dsDNA hybrids, incubation of the mixture at 30 to 50°C below the $T_m$ will allow hybrids of more diverse ssDNAs to form. Incubation at 10 to 15°C below the $T_m$ permits hybrid formation only with almost identical strands.

In one of the more widely used hybridization techniques, nitrocellulose filters with bound nonradioactive DNA strands are incubated at the appropriate temperature with single-stranded DNA fragments made radioactive with $^{32}P$, $^{3}H$, or $^{14}C$. After radioactive fragments are allowed to hybridize with the membrane-bound ssDNA, the membrane is washed to remove any nonhybridized ssDNA and its radioactivity is measured. The quantity of radioactivity bound to the filter reflects the amount of hybridization and thus the similarity of the DNA sequences. The degree of similarity or homology is expressed as the percent of experimental DNA radioactivity retained on the filter compared with the percent of homologous DNA radioactivity bound under the same conditions (**table 19.6** provides examples). Two strains whose DNAs show at least 70% relatedness under optimal hybridization conditions and less than a 5% difference in $T_m$ often are considered members of the same species.

If DNA molecules are very different in sequence, they will not form a stable, detectable hybrid. Therefore DNA-DNA hybridization is used to study only closely related microorganisms. More distantly related organisms are compared by carrying out DNA-RNA hybridization experiments using radioactive ribosomal or transfer RNA. Distant relationships can be detected because rRNA

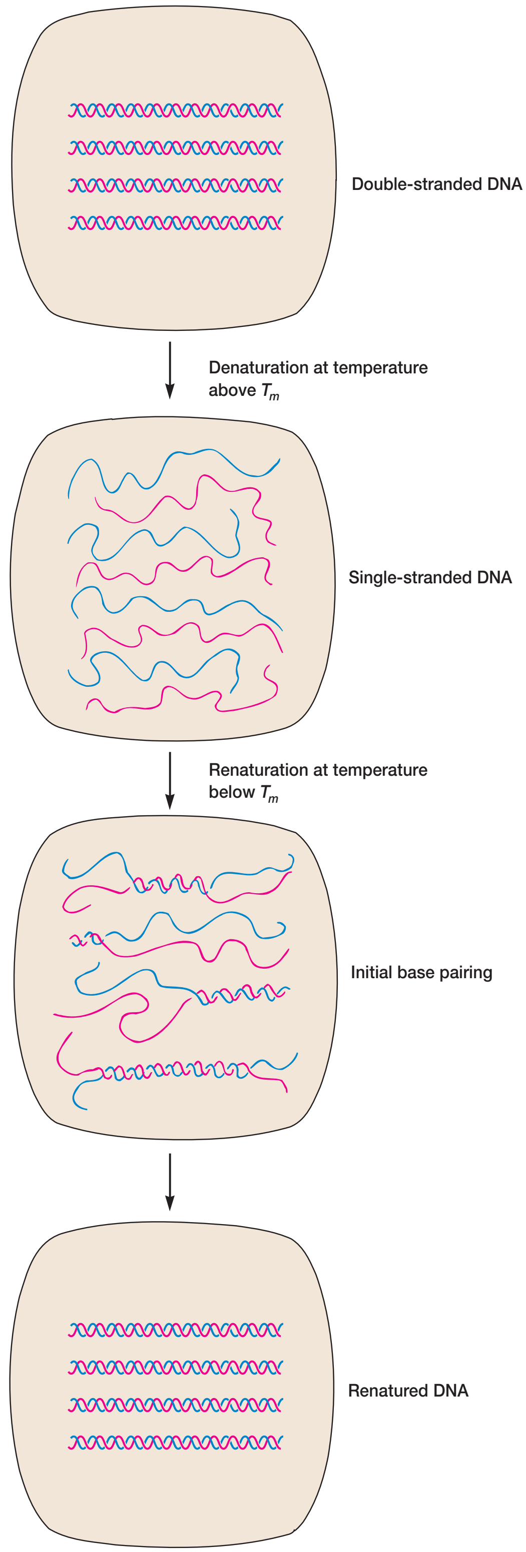



**Figure 19.7 Nucleic Acid Melting and Hybridization.** Complementary strands are shown in different colors.

**Table 19.6 Comparison of *Neisseria* Species by DNA Hybridization Experiments**

| Membrane-Attached DNA[a] | Percent Homology[b] |
|---|---|
| *Neisseria meningitidis* | 100 |
| *N. gonorrhoeae* | 78 |
| *N. sicca* | 45 |
| *N. flava* | 35 |

Source: Data from T. E. Staley and R. R. Colwell, "Applications of Molecular Genetics and Numerical Taxonomy to the Classification of Bacteria" in *Annual Review of Ecology and Systematics,* 8: 282, 1973.

[a]The experimental membrane-attached nonradioactive DNA from each species was incubated with radioactive *N. meningitidis* DNA, and the amount of radioactivity bound to the membrane was measured. The more radioactivity bound, the greater the homology between DNA sequences.

[b] $\frac{\textit{N. meningitidis}\text{ DNA bound to experimental DNA}}{\text{Amount bound to membrane-attached }\textit{N. meningitidis}\text{ DNA}} \times 100$

and tRNA genes represent only a small portion of the total DNA genome and have not evolved as rapidly as most other microbial genes. The technique is similar to that employed for DNA-DNA hybridization: membrane-bound DNA is incubated with radioactive rRNA, washed, and counted. An even more accurate measurement of homology is obtained by finding the temperature required to dissociate and remove half the radioactive rRNA from the membrane; the higher this temperature, the stronger the rRNA-DNA complex and the more similar the sequences. Ribosomes and ribosomal RNA (pp. 267–68); Transfer RNA (pp. 266–67)

### Nucleic Acid Sequencing

Despite the usefulness of G + C content determination and nucleic acid hybridization studies, genome structures can be directly compared only by sequencing DNA and RNA. Techniques for rapidly sequencing both DNA and RNA are now available; thus far RNA sequencing has been used more extensively in microbial taxonomy.

Most attention has been given to sequences of the 5S and 16S rRNAs isolated from the 50S and 30S subunits, respectively, of procaryotic ribosomes (*see sections 3.3 and 12.2*). The rRNAs are almost ideal for studies of microbial evolution and relatedness since they are essential to a critical organelle found in all microorganisms. Their functional role is the same in all ribosomes. Furthermore, their structure changes very slowly with time, presumably because of their constant and critical role. Because rRNA contains variable and stable sequences, both closely related and very distantly related microorganisms can be compared. This is an important advantage as distantly related organisms can be studied only using sequences that change little with time.

There are several ways to sequence rRNA. Ribosomal RNAs can be characterized in terms of partial sequences by the oligonucleotide cataloging method as follows. Purified, radioactive 16S rRNA is treated with the enzyme $T_1$ ribonuclease, which cleaves it into fragments. The fragments are separated, and all fragments composed of at least six nucleotides are sequenced. The sequences of corresponding 16S rRNA fragments from different procaryotes are then aligned and compared using a computer, and association coefficients ($S_{ab}$ values) are calculated. Complete rRNAs now are sequenced using procedures like the following. First, RNA is isolated and purified. Then, reverse transcriptase is used to make complementary DNA (cDNA) using primers that are complementary to conserved rRNA sequences. Next, the polymerase chain reaction amplifies the cDNA. Finally, the cDNA is sequenced and the rRNA sequence deduced from the results. The polymerase chain reaction (pp. 326–27); DNA sequencing (pp. 345–47)

Recently complete procaryotic genomes have been sequenced (*see chapter 15*). Direct comparison of complete genome sequences undoubtedly will become important in procaryotic taxonomy.

1. Summarize the advantages of using each major group of characteristics (morphological, physiological/metabolic, ecological, genetic, and molecular) in classification and identification. How is each group related to the nature and expression of the genome? Give examples of each type of characteristic.
2. What two modes of genetic exchange in procaryotes have proved taxonomically useful? Why are plasmids of such importance in bacterial taxonomy?
3. Briefly describe some ways in which proteins from different organisms can be compared.
4. What is the G + C content of DNA, and how can it be determined through melting temperature studies and density gradient centrifugation?
5. Discuss the use of G + C content in taxonomy. Why is it not safe to assume that two microorganisms with the same G + C content belong to the same species? In what two ways are G + C content data taxonomically valuable?
6. Describe how nucleic acid hybridization studies are carried out using membrane-bound DNA. Why might one wish to vary the incubation temperature during hybridization? What is the advantage of conducting DNA-RNA hybridization studies?
7. How are rRNA sequencing studies conducted, and why is rRNA so suitable for determining relatedness?

## 19.6 Assessing Microbial Phylogeny

Procaryotic taxonomy is changing rapidly. This is caused by ever-increasing knowledge of the biology of procaryotes and remarkable advances in computers and the use of molecular characteristics to determine phylogenetic relationships between procaryotic groups. This section briefly describes some of the ways in which phylogenetic relationships are determined.

### Molecular Chronometers

The sequences of nucleic acids and proteins change with time and are considered to be **molecular chronometers.** This concept, first suggested by Zuckerkandl and Pauling (1965), is important in the use of molecular sequences in determining phylogenetic relationships and is based on the assumption that there is an evolutionary clock. It is thought that the sequences of many rRNAs and proteins gradually change over time without destroying or severely altering their functions. One assumes that such changes are selectively neutral, occur fairly randomly, and increase linearly with time. When the sequences of similar molecules are quite different in two groups of organisms, the groups diverged from one another a long time ago. Phylogenetic analysis using molecular chronometers is somewhat complex because the rate of sequence change can vary; some periods are characterized by especially rapid change. Furthermore, different molecules and various parts of the same molecule can change at different rates. Highly conserved molecules such as rRNAs are used to follow large-scale evolutionary changes, whereas rapidly changing molecules are employed in following speciation. Not everyone believes that molecular chronometers, and particularly protein clocks, are very accurate. Further studies will be required to establish their accuracy and usefulness.

### Phylogenetic Trees

Phylogenetic relationships are illustrated in the form of branched diagrams or trees. A **phylogenetic tree** is a graph made of branches that connect nodes (**figure 19.8**). The nodes represent taxonomic units such as species or genes; the external nodes, those at the end of the branches, represent living organisms. The tree may have a time scale, or the length of the branches may represent the number of molecular changes that have taken place between the two nodes. Finally, a tree may be unrooted or rooted. An unrooted tree (figure 19.8*a*) simply represents phylogenetic relationships but does not provide an evolutionary path. Figure 19.8*a* shows that A is more closely related to C than it is to either B or D, but does not specify the common ancestor for the four species or the direction of change. In contrast, the rooted tree (figure 19.8*b*) does give a node that serves as the common ancestor and shows the development of the four species from this root. It is much more difficult to develop a rooted tree. For example, there are 15 possible rooted trees that connect four species, but only three possible unrooted trees.

Phylogenetic trees are developed by comparing molecular sequences. To compare two molecules their sequences must first be aligned so that similar parts match up. The object is to align and compare homologous sequences, ones that are similar because they had a common origin in the past. This is not an easy task, and computers plus fairly complex mathematics must be employed to minimize the number of gaps and mismatches in the sequences being compared.

Once the molecules have been aligned, the number of positions that vary in the sequences can be determined. These data are used to calculate a measure of the difference between the sequences. Often the difference is expressed as the **evolutionary distance.** This is simply a quantitative indication of the number of positions that differ between two aligned macromolecules. Statistical adjustments can be made for back mutations and multiple substitutions that may have occurred. Organisms are then clustered together based on similarity in the sequences. The most similar organisms are clustered together, then compared with the remaining organisms to form a larger cluster associated together at a lower level of similarity or evolutionary distance. The process continues until all organisms are included in the tree.

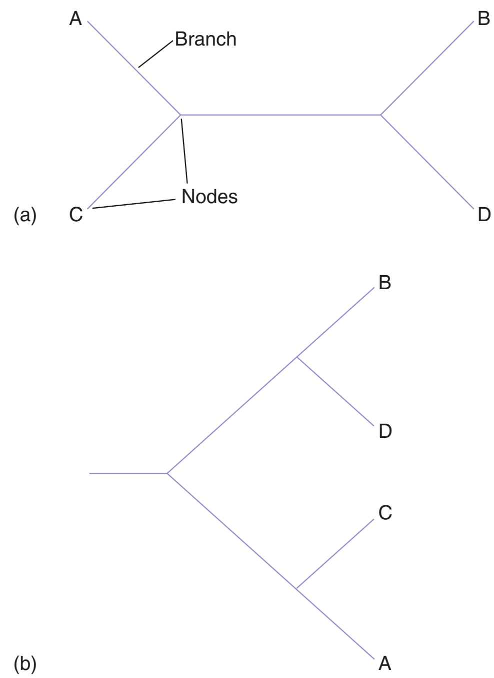



**Figure 19.8 Examples of Phylogenetic Trees.** (**a**) Unrooted tree joining four taxonomic units. (**b**) Rooted tree. See text for details.

Phylogenetic relationships also can be estimated by techniques such as parsimony analysis. In this approach, relationships are determined by estimating the minimum number of sequence changes required to give the final sequences being compared. It is presumed that evolutionary change occurs along the shortest pathway with the fewest changes or steps from an ancestor to the organism in question. The tree or pattern of relationships is favored that is simplest and requires the fewest assumptions.

### rRNA, DNA, and Proteins as Indicators of Phylogeny

Although a variety of molecular techniques are used in estimating the phylogenetic relatedness of procaryotes, the comparison of 16S rRNAs isolated from thousands of strains is of particular

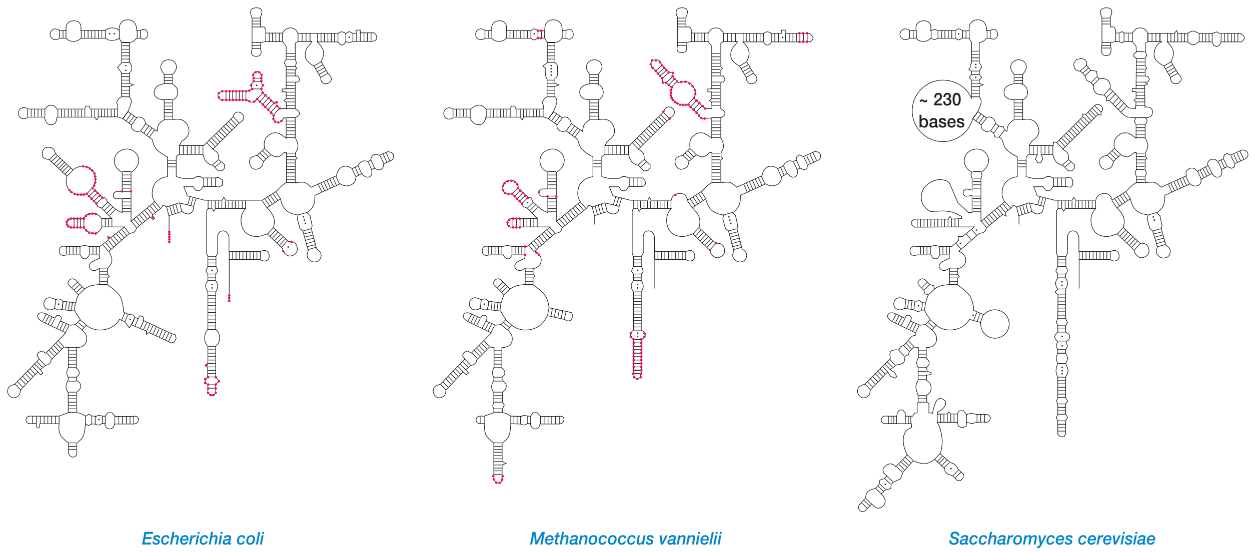



**Figure 19.9 Small Ribosomal Subunit RNA.** Representative examples of rRNA secondary structures from the three primary domains: bacteria (*Escherichia coli*), archaea (*Methanococcus vannielii*), and eucarya (*Saccharomyces cerevisiae*). The red dots mark positions where bacteria and archaea normally differ. *Source: Data from C. P. Woese.* Microbiological Reviews, *51(2):221–227, 1987.*

importance (**figure 19.9**). Recall that either complete rRNA or rRNA fragments can be sequenced and compared (p. 432). The association coefficients or $S_{ab}$ values from rRNA studies are assumed to be a true measure of relatedness; the higher the $S_{ab}$ values obtained from comparing two organisms, the more closely the organisms are related to each other. If the sequences of the 16S rRNAs of two organisms are identical, the $S_{ab}$ value is 1.0. $S_{ab}$ values also are a measure of evolutionary time. A group of procaryotes that branched off from other procaryotes long ago will exhibit a large range of $S_{ab}$ values because it has had more time to diversify than a group that developed more recently. That is, the narrower the range of $S_{ab}$ values in a group of procaryotes, the more modern it is. After $S_{ab}$ values have been determined, a computer calculates the relatedness of the organisms and summarizes their relationships in a tree or dendrogram (figures 19.5 and 19.8).

Ribosomal RNA sequence studies have uncovered a feature of great practical importance. The 16S rRNA of most major phylogenetic groups has one or more characteristic nucleotide sequences called oligonucleotide signatures. **Oligonucleotide signature sequences** are specific oligonucleotide sequences that occur in most or all members of a particular phylogenetic group. They are rarely or never present in other groups, even closely related ones. Thus signature sequences can be used to place microorganisms in the proper group. Signature sequences have been identified for bacteria, archaea, eucaryotes, and many major procaryotic groups (**table 19.7**).

**Table 19.7** Selected 16S rRNA Signature Sequences for Some Bacterial Groups[a]

| Position in rRNA | Consensus Composition | γ-Proteobacteria | Cyanobacteria | Spirochetes | *Bacteroides* | Green Sulfur | Green Nonsulfur | *Deinococcus* | Gram Positive (Low GC) | Gram Positive (High GC) | *Planctomyces* |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 47 | C | + | + | U | + | + | + | + | + | + | G |
| 53 | A | + | + | G | + | + | G | + | + | + | G |
| 570 | G | + | + | + | U | + | + | + | + | + | U |
| 812 | G | c | + | + | + | + | + | C | + | + | + |
| 906 | G | Ag | + | + | + | + | A | + | + | A | + |
| 955 | U | + | + | + | + | + | + | + | + | AC | C |
| 1,207 | G | + | C | + | + | + | + | + | C | C | + |
| 1,234 | C | + | + | a | U | A | + | + | + | + | + |

[a]A plus sign in a column means that the group has the same base as the consensus sequence. If the letter is given in upper case, it is changed in more than 90% of the cases. A lowercase letter signifies a minor occurrence base (<15% of the cases).

Although rRNA comparisons are useful above the species level, DNA similarity studies sometimes are more effective in categorizing individual species and genera. These comparisons can be carried out using G + C content or hybridization studies, as already discussed. Techniques such as direct sequence analysis and analysis of DNA restriction fragment patterns also can be used. There are advantages to DNA comparisons. As with rRNA, the DNA composition of a cell does not change with growth conditions. DNA comparisons also are based on the complete genomes rather than a fraction, and make it easier to precisely define a species based on the 70% relatedness criterion. Full sequences of genomes now are being published and will make it easier to study the impact on phylogenetic schemes of such processes as lateral gene transfer as will be discussed later.

Many protein sequences are currently used to develop phylogenetic trees. This approach does have some advantages over rRNA comparisons. A sequence of 20 amino acids has more information per site than a sequence of four nucleotides. Protein sequences are less affected by organism-specific differences in G + C content than are DNA and RNA sequences. Finally, protein sequence alignment is easier because it is not dependent on secondary structure as is an rRNA sequence. Proteins evolve at different rates, as might be expected. Indispensable proteins with constant functions do not change as rapidly (e.g., histones and heat-shock proteins), whereas proteins such as immunoglobulins evolve quite rapidly. Thus not all proteins are suitable for studying large-scale changes that occur over long periods. As mentioned earlier, there is a question about the adequacy of protein-based clocks.

It is clear that sequences of all three macromolecules can provide valuable phylogenetic information. However, different sequences sometimes produce different trees, and it may be difficult to decide which result is most accurate. Presumably more molecular data plus further study of phenotypic properties will help resolve uncertainties.

### Polyphasic Taxonomy

Because phylogenetic results vary with the data used in analysis, many taxonomists believe that all possible valid data should be employed in determining phylogeny. In the approach called **polyphasic taxonomy,** taxonomic schemes are developed using a wide range of phenotypic and genotypic information ranging from molecular properties to ecological characteristics. The techniques that are appropriate for grouping organisms depend on the level of taxonomic resolution needed. For example, serological techniques can be used to identify strains, but not genera or species. Protein electrophoretic patterns are useful in determining species, but not genera or families. DNA hybridization and the analysis of % G + C can be used in studying species and genera. Characteristics such as chemical composition, DNA probe results, rRNA sequences, and DNA sequences can be used to define species, genera, and families. Where possible, as many properties as possible are used to get more stable and reliable results. Successful polyphasic approaches often will help one select techniques for rapid identification of the microorganism.

Because rRNA sequences have been used so extensively, we will focus mainly on the phylogenetic trees derived from rRNA studies.

1. What are molecular chronometers and upon what assumptions are they based?
2. Define phylogenetic tree and evolutionary distance. What is the difference between an unrooted and a rooted tree?
3. Discuss the use of $S_{ab}$ values in determining the relatedness of procaryotes and the evolutionary age of taxonomic groups. What are oligonucleotide signature sequences?
4. Why might one choose to use DNA or protein sequences in phylogenetic studies?
5. Describe polyphasic taxonomy and discuss some of its advantages.

## 19.7 The Major Divisions of Life

Since the beginning of biology, organisms have been classified as either plants or animals. However, discoveries in microbiology over the past century have shown that the two-kingdom system is oversimplified. Although not all biologists would agree, most microbiologists now believe that living forms can be divided into three distinctly different groups. We will first review this system in more detail, then turn to alternate views.

### Domains

As mentioned earlier and illustrated in figure 19.3 (p. 424), Carl Woese and his collaborators have used rRNA studies to group all living organisms into three domains: *Archaea, Bacteria,* and *Eucarya.* Thus there are two quite different groups of procaryotes, the bacteria and the archaea. The bacteria comprise the vast majority of procaryotes. Among other properties, bacteria either have cell wall peptidoglycan containing muramic acid or are related to bacteria with such cell walls, and have membrane lipids with ester-linked, straight-chained fatty acids that resemble eucaryotic membrane lipids (**table 19.8**). The second group, the archaea differ from bacteria in many respects and resemble eucaryotes in some ways (table 19.8). Although the Archaea are described in more detail at a later point, it should be noted that they differ from bacteria in lacking muramic acid in their cell walls and in possessing (1) membrane lipids with ether-linked branched aliphatic chains, (2) transfer RNAs without thymidine in the T or TψC arm (*see section 12.2*), (3) distinctive RNA polymerase enzymes, and (4) ribosomes of different composition and shape. Thus although archaea resemble bacteria in their procaryotic cell structure, they vary considerably on the molecular level. Both groups differ from eucaryotes in their cell ultrastructure (*see pp. 91–92)* and many other properties. However, inspection of table 19.8 shows that both bacteria and archaea do share some biochemical properties with eucaryotic cells. For example, bacteria and eucaryotes have ester-linked membrane lipids; archaea and eucaryotes are similar with respect to some components of the RNA and protein synthetic systems. *The archaea (chapter 20)*

Although the preceding view is the most widely accepted, other phylogenetic trees have been proposed. Six or more different trees relating the major domains have been proposed.

**Table 19.8** Comparison of *Bacteria, Archaea,* and *Eucarya*

| Property | *Bacteria* | *Archaea* | *Eucarya* |
|---|---|---|---|
| **Membrane-Enclosed Nucleus with Nucleolus** | Absent | Absent | Present |
| **Complex Internal Membranous Organelles** | Absent | Absent | Present |
| **Cell Wall** | Almost always have peptidoglycan containing muramic acid | Variety of types, no muramic acid | No muramic acid |
| **Membrane Lipid** | Have ester-linked, straight-chained fatty acids | Have ether-linked, branched aliphatic chains | Have ester-linked, straight-chained fatty acids |
| **Gas Vesicles** | Present | Present | Absent |
| **Transfer RNA** | Thymine present in most tRNAs | No thymine in T or TψC arm of tRNA | Thymine present |
| | *N*-formylmethionine carried by initiator tRNA | Methionine carried by initiator tRNA | Methionine carried by initiator tRNA |
| **Polycistronic mRNA** | Present | Present | Absent |
| **mRNA Introns** | Absent | Absent | Present |
| **mRNA Splicing, Capping, and Poly A Tailing** | Absent | Absent | Present |
| **Ribosomes** | | | |
| Size | 70S | 70S | 80S (cytoplasmic ribosomes) |
| Elongation factor 2 | Does not react with diphtheria toxin | Reacts | Reacts |
| Sensitivity to chloramphenicol and kanamycin | Sensitive | Insensitive | Insensitive |
| Sensitivity to anisomycin | Insensitive | Sensitive | Sensitive |
| **DNA-Dependent RNA Polymerase** | | | |
| Number of enzymes | One | Several | Three |
| Structure | Simple subunit pattern (4 subunits) | Complex subunit pattern similar to eucaryotic enzymes (8–12 subunits) | Complex subunit pattern (12–14 subunits) |
| Rifampicin sensitivity | Sensitive | Insensitive | Insensitive |
| **Polymerase II Type Promoters** | Absent | Present | Present |
| **Metabolism** | | | |
| Similar ATPase | No | Yes | Yes |
| Methanogenesis | Absent | Present | Absent |
| Nitrogen fixation | Present | Present | Absent |
| Chlorophyll-based photosynthesis | Present | Absent | Present[a] |
| Chemolithotrophy | Present | Present | Absent |

a Present in chloroplasts (of bacterial origin).

**Figure 19.10** provides a simplified view of some of these. The first (figure 19.10*a*) indicates that the three groups are about equidistant from one another and fits with the early rRNA data. Figure 19.10*b* represents the currently most popular tree in which archaea and eucaryotes have a common ancestor; organisms like the bacteria may have existed before the other domains. The third tree, called the eocyte tree (figure 19.10*c*), is based on the proposal that sulfur-dependent, extremely thermophilic procaryotes called eocytes (dawn + cell) are a separate group and more closely related to eucaryotes than are the archaea. Finally, some have proposed that eucaryotic cells are chimeric and arose from the fusion of a bacterium and archaeon (possibly a bacterium lacking a cell wall engulfed an eocytelike archaeon) (figure 19.10*d*).

Clearly the situation is confused and more than one model has been proposed, though most microbiologists favor the three-domain tree in figure 19.10*b*. When some protein sequences are used to construct phylogenetic trees, one does not even get a three-domain pattern. Many factors may account for these problems. There could be unrecognized gene duplications that occurred before the domains formed, leading to confusing patterns. Unequal rates of evolution could distort the trees. Phylogenetically important information may have been lost in some molecular sequences. There may be significant sequence variation between the same molecules from different strains of the same species. Unless several strains are analyzed, false conclusions may be drawn. Thus inaccurate universal trees may result when only the sequences from a few molecules are employed (as is usually the case).

One of the most important difficulties in constructing a satisfactory tree is widespread, frequent horizontal or lateral gene transfer. Recent genome sequence studies have shown that there is extensive horizontal gene transfer within and between domains (*see pp. 352–53*). Eucaryotes possess genes from both bacteria and archaea, and there has been frequent gene swapping between the two procaryotic domains. It appears that at least some bacteria even have acquired eucaryotic genes. Thus the pattern of microbial evolution is not as linear and treelike as previously thought. The trees shown in figures 19.3 and 19.10 are undoubtedly oversimplified. **Figure 19.11** depicts a more

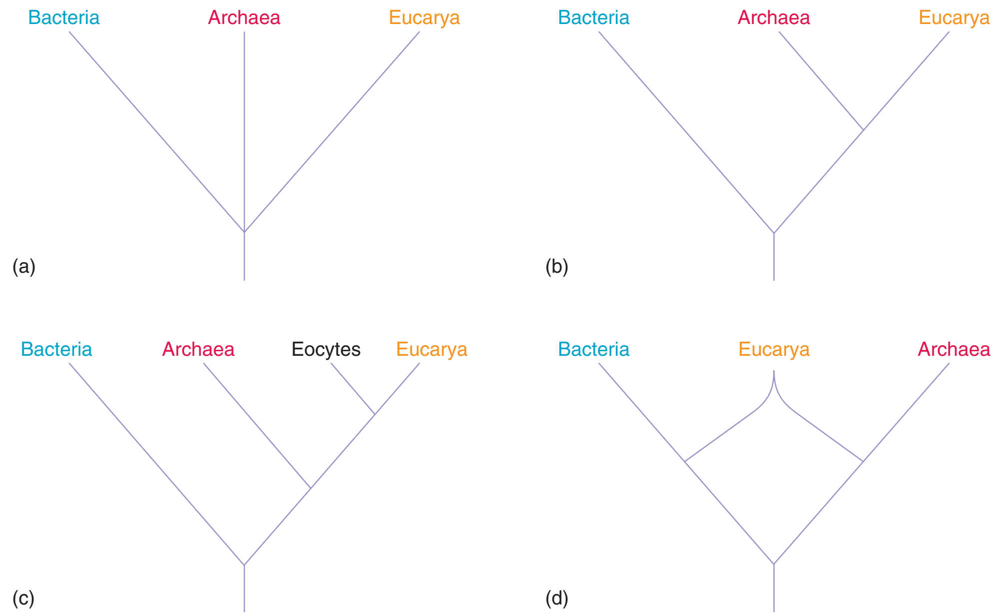



**Figure 19.10 Variations in the Design of the "Tree of Life."** These four alternative phylogenetic trees are discussed in the text.

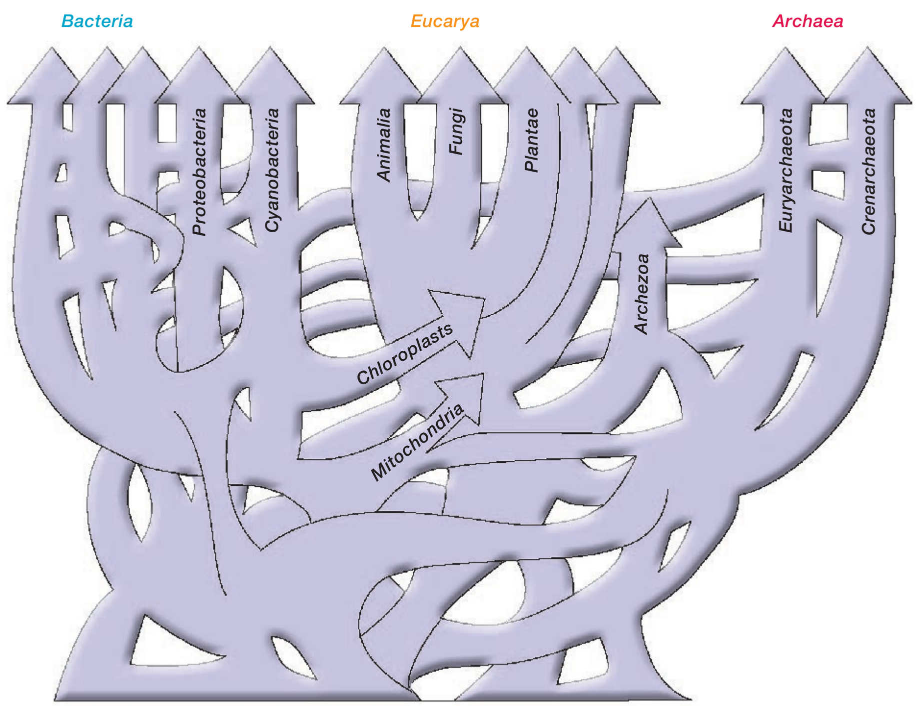



**Figure 19.11 Universal Phylogenetic Tree with Frequent Horizontal or Lateral Gene Transfers.** See text for discussion.

realistic reticulated tree in which horizontal gene transfer plays a major role. This tree resembles a web or network with many lateral branches linking various trunks, each branch representing the transfer of one or a few genes. Instead of having a single main trunk or common ancestor at its base, this tree has several trunks or groups of primitive cells that contribute to the original gene pool. Although there is extensive gene transfer between the two procaryotic domains throughout their development, the eucaryotic domain seldom participates in horizontal gene transfer after the formation of fungi, plants, and animals. It is possible that eucaryotic cells originated in a complex process involving many gene transfers from both bacteria and archaea. This hypothesis still allows for the formation of mitochondria and chloroplasts by endosymbiosis with α-proteobacteria and cyanobacteria, respectively. Presumably the three domains remain separate because there are many more gene transfers within each than between domains.

This brief discussion of the problems in developing a true universal phylogenetic tree is intended to show the difficulty in determining phylogenetic relationships. The best results will be obtained when all possible data, both molecular and phenotypic, are used in the analysis (for example, in polyphasic taxonomy). We will usually employ trees derived from 16S rRNA sequences because these data are most extensive and are used by most microbiologists. Keep in mind that such trees may well change as further data are collected and analyzed.

## Kingdoms

While most bacteriologists favor the three-domain system, many protozoologists, botanists, and zoologists still think in terms of five or more kingdoms. This section briefly summarizes the nature of some of these classification systems.

The first classification system to have gained popularity in the last few decades is the five-kingdom system first suggested by Robert H. Whittaker in the 1960s. An overview of Whittaker's five-kingdom system is presented in **figure 19.12*a*.** Organisms are placed into five kingdoms based on at least three major criteria: (1) cell type—procaryotic or eucaryotic, (2) level of organization—solitary and colonial unicellular organization or multicellular, and (3) nutritional type. In this system the kingdom *Animalia* contains multicellular animals with wall-less eucaryotic cells and primarily ingestive nutrition, whereas the kingdom *Plantae* is composed of multicellular plants with walled eucaryotic cells and primarily photoautotrophic nutrition. Microbiologists study members of the other three kingdoms. The kingdom *Monera* or *Procaryotae* contains all procaryotic organisms. The kingdom *Protista* is the least homogeneous and hardest to define. **Protists** are eucaryotes with unicellular organization, either in the form of solitary cells or colonies of cells lacking true tissues. They may have ingestive, absorptive, or photoautotrophic nutrition, and they include most of the microorganisms known as algae, protozoa, and many of the simpler fungi. The kingdom *Fungi* contains eucaryotic and predominately multinucleate organisms, with nuclei dispersed in a walled and often septate mycelium (*see chapter 25*); their nutrition is absorptive. The taxonomy of the major protist and fungal phyla is discussed in more detail in chapters 25 to 27.

The five-kingdom system is not accepted by many biologists. A major problem is its lack of distinction between archaea and bacteria. The kingdom *Protista* also may be too diverse to be taxonomically useful. In addition, the boundaries between the kingdoms *Protista, Plantae,* and *Fungi* are ill-defined. For example, the brown algae are probably not closely related to the plants even though the five-kingdom system places them in the *Plantae.*

Because of such problems with the five-kingdom system, various alternatives have been suggested. The six-kingdom system is the simplest option; it divides the kingdom *Monera* or *Procaryotae* into two kingdoms, the *Eubacteria* and *Archaeobacteria* (figure 19.12*b*). Many attempts have been made to divide the protists into several better-defined kingdoms. The eight-kingdom system of Cavalier-Smith is a good example (figure 19.12*c*). Cavalier-Smith believes that differences in cellular structure and genetic organization are exceptionally important in determining phylogeny; thus he has used ultrastructural characteristics as well as rRNA sequences and other molecular data in developing his classification. He divides all organisms into two empires and eight kingdoms. The empire *Bacteria* contains two kingdoms, the *Eubacteria* and the *Archaeobacteria.* The second empire, the *Eucaryota,* contains six kingdoms of eucaryotic organisms. There are two new kingdoms of eucaryotes. The *Archezoa* are primitive eucaryotic unicellular organisms such as *Giardia* that have 70S ribosomes and lack Golgi apparatuses, mitochondria, chloroplasts, and peroxisomes. The kingdom *Chromista* contains mainly photosynthetic organisms that have their chloroplasts within the lumen of the rough endoplasmic reticulum rather than in the cytoplasmic matrix (as is the case in the kingdom *Plantae*). Diatoms, brown algae, cryptomonads, and oomycetes are all placed in the *Chromista.* The boundaries of the remaining four kingdoms—*Plantae, Fungi, Animalia,* and *Protozoa*—have been adjusted to better define each kingdom and distinguish it from the others. Sogin and his coworkers do not cluster the eucaryotes into a few major divisions, but rather consider them to be a single domain or empire composed of a collection of independently evolved lineages (figure 19.12*d*). In this scheme the protists do not comprise a separate kingdom, but simply represent a level of organization with many separate lineages and tremendous diversity.

1. How does Woese divide organisms into domains or empires in his universal phylogenetic tree? Describe several of the major differentiating characteristics of each domain.
2. Describe the two major alternatives to Woese's tree that are depicted in figure 19.10*c* and 19.10*d*. Why have there been difficulties in developing an accurate tree? Discuss the effect of frequent horizontal gene transfer on phylogenetic trees.
3. With what three major criteria did Whittaker divide organisms into five kingdoms?
4. Give the names and main distinguishing characteristics of the five kingdoms.
5. Briefly describe the six- and eight-kingdom systems. How do they differ from the five-kingdom system?

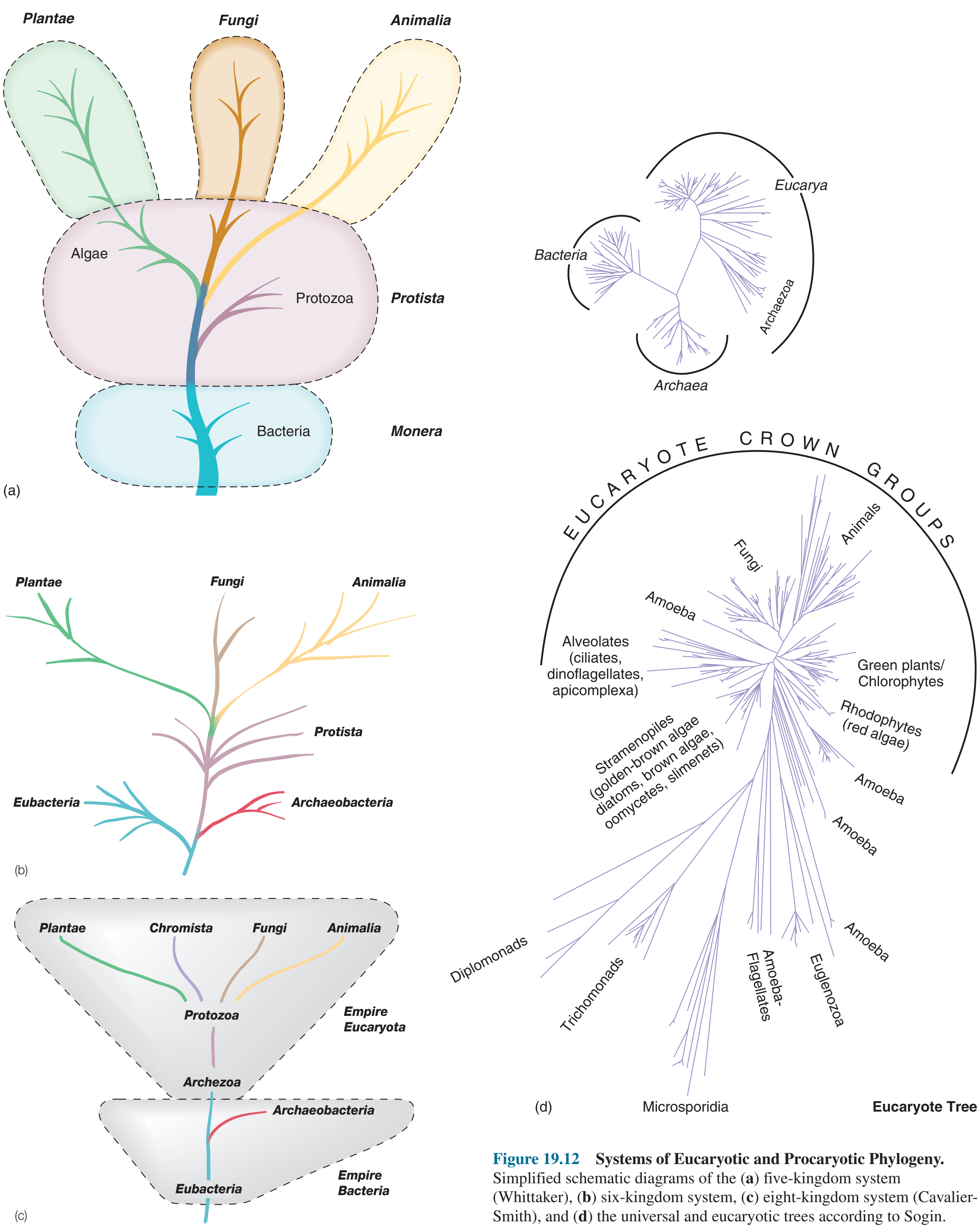



**Figure 19.12 Systems of Eucaryotic and Procaryotic Phylogeny.** Simplified schematic diagrams of the (**a**) five-kingdom system (Whittaker), (**b**) six-kingdom system, (**c**) eight-kingdom system (Cavalier-Smith), and (**d**) the universal and eucaryotic trees according to Sogin.

## Box 19.1

### "Official" Nomenclature Lists: A Letter from Bergey's*

On a number of occasions lately, the impression has been given that the status of a bacterial taxon in *Bergey's Manual of Systematic Bacteriology* or *Bergey's Manual of Determinative Bacteriology* is in some sense official. Similar impressions are frequently given about the status of names in the *Approved List of Bacterial Names* and in the Validation Lists of newly proposed names that appear regularly in the *International Journal of Systematic Bacteriology.* It is therefore important to clarify these matters.

There is no such thing as an official classification. *Bergey's Manual* is not "official"—it is merely the best consensus at the time, and although great care has always been taken to obtain a sound and balanced view, there are also always regions in which data are lacking or confusing, resulting in differing opinions and taxonomic instability. When *Bergey's Manual* disavows that it is an official classification, many bacteriologists may feel that the solid earth is trembling. But many areas are in fact reasonably well established. Yet taxonomy is partly a matter of judgment and opinion, as is all science, and until new information is available, different bacteriologists may legitimately hold different views. They cannot be forced to agree to any "official classification." It must be remembered that, as yet, we know only a small percentage of the bacterial species in nature. Advances in technique also reveal new lights on bacterial relationships. Thus we must expect that existing boundaries of groups will have to be redrawn in the future, and it is expected that molecular biology, in particular, will imply a good deal of change over the next few decades.

The position with the Approved Lists and the Validation Lists is rather similar. When bacteriologists agreed to make a new start in bacteriological nomenclature, they were faced with tens of thousands of names in the literature of the past. The great majority were useless, because, except for about 2,500 names, it was impossible to tell exactly what bacteria they referred to. These 2,500 were therefore retained in the Approved Lists. The names are only approved in the sense that they were approved for retention in the new bacteriological nomenclature. The remainder lost standing in the nomenclature, which means they do not have to be considered when proposing new bacterial names (although names can be individually revived for good cause under special provisions).

The new International Code of Nomenclature of Bacteria requires all new names to be validly published to gain standing in the nomenclature, either by being published in papers in the *International Journal of Systematic Bacteriology* or, if published elsewhere, by being announced in the Validation Lists. The names in the Validation Lists are therefore valid only in the sense of being validly published (and therefore they must be taken account of in bacterial nomenclature). The names do not have to be adopted in all circumstances; if users believe the scientific case for the new taxa and validly published names is not strong enough, they need not adopt the names. For example, *Helicobacter pylori* was immediately accepted as a replacement for *Campylobacter pylori* by the scientific community, whereas *Tatlockia micdadei* had not generally been accepted as a replacement for *Legionella micdadei.* Taxonomy remains a matter of scientific judgment and general agreement.

*From P. H. A. Sneath and D. J. Brenner, "Official Nomenclature Lists in *ASM News,* 58(4):175, 1992. Copyright © by the American Society for Microbiology. Reprinted by permission.

## 19.8 Bergey's Manual of Systematic Bacteriology

In 1923, David Bergey, professor of bacteriology at the University of Pennsylvania, and four colleagues published a classification of bacteria that could be used for identification of bacterial species, the *Bergey's Manual of Determinative Bacteriology.* This manual is now in its ninth edition. The first edition of *Bergey's Manual of Systematic Bacteriology,* a more detailed work that contains descriptions of all procaryotic species currently identified, also is available (**Box 19.1**). The first volume of the second edition has been published recently. This section briefly describes the current edition of *Bergey's Manual of Systematic Bacteriology* (or *Bergey's Manual*) and then discusses at more length the new second edition.

### The First Edition of *Bergey's Manual of Systematic Bacteriology*

Because it has not been possible in the past to classify procaryotes satisfactorily based on phylogenetic relationships, the system given in the first edition of *Bergey's Manual of Systematic Bacteriology* is primarily phenetic. Each of the 33 sections in the four volumes contains procaryotes that share a few easily determined characteristics and bears a title that either describes these properties or provides the vernacular names of the procaryotes included. The characteristics used to define sections are normally features such as general shape and morphology, Gram-staining properties, oxygen relationship, motility, the presence of endospores, the mode of energy production, and so forth. Procaryotic groups are divided among the four volumes in the following manner: (1) gram-negative bacteria of general, medical, or industrial importance; (2) gram-positive bacteria other than actinomycetes; (3) gram-negative bacteria with distinctive properties, cyanobacteria, and archaea; and (4) actinomycetes (gram-positive filamentous bacteria).

Gram-staining properties play a singularly important role in this phenetic classification; they even determine the volume into which a species is placed. There are good reasons for this significance. As noted in chapter 3, Gram staining usually reflects fundamental differences in bacterial wall structure. Gram-staining properties also are correlated with many other properties of bacteria. Typical gram-negative bacteria, gram-positive bacteria, and mycoplasmas (bacteria lacking walls) differ in many characteristics, as can be seen in **table 19.9.** For these and other reasons, bacteria traditionally have been classified as gram positive or gram negative. This approach is retained to some extent in more phylogenetic classifications and is a useful way to think about bacterial diversity.

The procaryotic cell wall (pp. 55–61)

**Table 19.9** Some Characteristic Differences between Gram-Negative and Gram-Positive Bacteria

| Property | Gram-negative Bacteria | Gram-positive Bacteria | Mycoplasmas |
|---|---|---|---|
| Cell wall | Gram-negative type wall with inner 2–7 nm peptidoglycan layer and outer membrane (7–8 nm thick) of lipid, protein, and lipopolysaccharide. (There may be a third outermost layer of protein.) | Gram-positive type wall with a homogeneous, thick cell wall (20–80 nm) composed mainly of peptidoglycan. Other polysaccharides and teichoic acids may be present. | Lack a cell wall and peptidoglycan precursors; enclosed by a plasma membrane |
| Cell shape | Spheres, ovals, straight or curved rods, helices or filaments; some have sheaths or capsules. | Spheres, rods, or filaments; may show true branching | Pleomorphic in shape; may be filamentous, can form branches |
| Reproduction | Binary fission, sometimes budding | Binary fission | Budding, fragmentation, and/or binary fission |
| Metabolism | Phototrophic, chemolithoautotrophic, or chemoorganoheterotrophic | Usually chemoorganoheterotrophic | Chemoorganoheterotrophic; most require cholesterol and long-chain fatty acids for growth. |
| Motility | Motile or nonmotile. Flagellation can be varied—polar, lophotrichous, peritrichous. Motility may also result from the use of axial filaments (spirochetes) or gliding motility. | Most often nonmotile; have peritrichous flagellation when motile | Usually nonmotile |
| Appendages | Can produce several types of appendages—pili and fimbriae, prosthecae, stalks | Usually lack appendages (may have spores on hyphae) | Lack appendages |
| Endospores | Cannot form endospores | Some groups can form endospores. | Cannot form endospores |

### The Second Edition of *Bergey's Manual of Systematic Bacteriology*

There has been enormous progress in procaryotic taxonomy since 1984, the year the first volume of *Bergey's Manual of Systematic Bacteriology* was published. In particular, the sequencing of rRNA, DNA, and proteins has made phylogenetic analysis of procaryotes feasible. As a consequence, the second edition of *Bergey's Manual* will be largely phylogenetic rather than phenetic and thus quite different from the first edition. Although the new edition will not be completed for some time, it is so important that its general features will be described here. Undoubtedly the details will change as work progresses, but the general organization of the new *Bergey's Manual* can be summarized.

The second edition will be published in five volumes. It will have more ecological information about individual taxa. The second edition will not group all the clinically important procaryotes together as the first edition did. Instead, pathogenic species will be placed phylogenetically and thus scattered throughout the following five volumes.

Volume 1—*The Archaea, and the Deeply Branching and Phototrophic Bacteria*

Volume 2—*The Proteobacteria*

Volume 3—*The Low G + C Gram-Positive Bacteria*

Volume 4—*The High G + C Gram-Positive Bacteria*

Volume 5—*The Planctomycetes, Spirochaetes, Fibrobacteres, Bacteroidetes,* and *Fusobacteria* (Volume 5 also will contain a section that updates descriptions and phylogenetic arrangements that have been revised since publication of volume 1.)

The second edition's five volumes will have a different organization than the first edition. The greatest change in organization of the volumes will be with respect to the gram-negative bacteria. The first edition describes all gram-negative bacteria in two volumes. Volume 1 contains the gram-negative bacteria of general, medical or industrial importance; volume 3 describes the archaea, cyanobacteria, and remaining gram-negative groups. The second edition describes the gram-negative bacteria in three volumes, with volume 2 reserved for the proteobacteria. The two editions treat the gram-positive bacteria more similarly. Although volume 2 of the first edition does have some high G + C bacteria, much of its coverage is equivalent to the new volume 3. Volume 4 of the first edition describes the actinomycetes and is similar to volume 4 of the second edition (high G + C gram-positive bacteria), although the new volume 4 will have broader coverage. For example, *Micrococcus* and *Corynebacterium* are in volume 2 of the first edition and will be in volume 4 of the second edition. **Table 19.10** summarizes the planned organization of the second edition and indicates where the discussion of a particular group may be found in this textbook. **Figure 19.13** depicts the major groups and their relatedness to each other. Bacterial classification according to *Bergey's Manual of Systematic Bacteriology* (appendices III and IV)

1. What characteristics are used to place procaryotes in different sections of *Bergey's Manual?*
2. What are the major ways in which gram-negative and gram-positive bacteria differ? Distinguish mycoplasmas from other bacteria.
3. Give several major ways in which the second edition of *Bergey's Manual* differs from the first edition.